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Watanabe's expansion: a solution for the convexity conundrum

D. GARCIA-LORITE^{*†‡} and R. MERINO[§]

[†]Quantitative Analyst Team, CaixaBank, Plaza de Castilla, 3, Madrid 28046, Spain

[‡]Facultat de Matemàtiques i Informàtica, Universitat de Barcelona, Gran Via 585, Barcelona 08007, Spain

[§]Market Risk Management Unit, VidaCaixa S.A., C/Juan Gris, 2-8, Barcelona 08014, Spain

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In this paper, we introduce a new method for pricing CMS derivatives. We utilize Malliavin's calculus to establish a model-free connection between the price of a CMS derivative and a quadratic payoff. Then, we apply Watanabe's expansions to quadratic payoffs under local and stochastic local volatility. The local and stochastic local volatility models are expressed in a general form, providing a generic approximation. To evaluate their accuracy, we will compare the approximations numerically under the normal SABR model against the market standards: Hagan's approximation and Monte Carlo simulation.

Keywords: Interest rates; Convexity adjustments; Watanabe expansion; Quadratic payoff; Local volatility; Local stochastic volatility

JEL Classification: G12, C58, C63

1. Introduction

Option pricing is one of the main tasks of a derivatives desk. Not only to obtain the derivatives prices and trade them, but also to calibrate more complex models or manage all the associated risks. In practice, most institutions consider stochastic volatility models to better capture market dynamics. However, including this feature adds more complexity to the calculations, making the processes more complicated and requiring more time to execute.

Over the last decades, has been aimed at developing various methods to accurately approximate option prices. Some of these methods focus on deriving expansions of implied volatility. For example, using perturbation methods as in Hagan and Woodward (1999), Hagan *et al.* (2002) and Jacquier and Lorig (2015), using heat kernel expansions as shown in Gatheral *et al.* (2012) or Henry-Labordère (2008), using asymptotic expansion methods as in Karami and Shiraya (2018), or Watanabe's expansion in Aït-Sahalia *et al.* (2021a, 2021b). Other approaches concentrate directly on approximating option prices. For example, the Alòs decomposition formula in Alòs (2012) expanded later in Alòs *et al.* (2020) and Gulisashvili *et al.* (2020), adjoint expansions as in Pagliarani and Pascucci (2013), the cos methods as in Fang and

Oosterlee (2009) or Watanabe's expansion in Kunitomo and Takahashi (2003), Takahashi and Yamada (2015, 2016), Takahashi *et al.* (2011, 2012), Li (2014) and Nagami (2021).

Despite the extensive literature on option approximations, few references exist on quadratic payoffs. Although these products are not commonly traded, they are useful for calculating the convexity correction for Constant Maturity Swaps (CMS). The main references are Hagan *et al.* (2019) and Hagan and Woodward (2021). The values of quadratic call options, put options, and swaps are defined as follows:

$$V^{QC}(T, F, K) := N_{T_p} \mathbb{E}^{\mathbb{Q}^N} [(F_T - K)_+^2], \quad (1)$$

$$V^{QP}(T, F, K) := N_{T_p} \mathbb{E}^{\mathbb{Q}^N} [(K - F_T)_+^2], \quad (2)$$

$$V^{QS}(T, F, K) := N_{T_p} \mathbb{E}^{\mathbb{Q}^N} [(F_T - K)^2]. \quad (3)$$

where T denotes the expiration date, T_p the payment date, and F_T the forward rate process associated with the derivative. Additionally, $\mathbb{E}^{\mathbb{Q}^N}[\cdot]$ represents the expectation with respect to the measure $\mathbb{Q}^N$, associated with the numeraire N_t . Although there is a closed formula for normal and log-normal models when volatility is constant, it is not good enough since these products depend more on market skews and smiles than vanilla options. Many banks use replication to price these structures.

*Corresponding author. Email: david.garcia.lorite@gmail.com

It is observed that the relationship

$$(F_T - K)_+^2 = 2 \int_K^\infty (F_T - K')_+ dK', \quad (4)$$

shows that the value of a quadratic call option can be precisely represented as the sum of the values of call options with varying strike prices. That is the main advantage of this approach, as it is consistent with vanilla option pricing, easy to implement, and automatically satisfies the put-call parity. Unfortunately, from a practical standpoint, a dense option grid with different strikes is necessary. Only a few options are traded per expiration and strike, so it is essential to calculate the grid, making it computationally intensive. In addition, the option prices with higher strikes, which are required for the calculation, are illiquid and have a 'fictitious' market price. To overcome this, an alternative methodology based on finding an approximation formula under the SABR model has been proposed by Hagan *et al.* (2019) and Hagan and Woodward (2021).

The paper aims to obtain an alternative approach for pricing CMS derivatives. Firstly, we will establish a connection between the price of a CMS derivative and a quadratic payoff using a bit of Malliavin calculus. This relationship is obtained without relying on any specific model. Then, we will extend the results of Watanabe's expansions to the case of quadratic payoffs over local and stochastic local volatility. The approximations are generic. To evaluate their accuracy, we will compare the approximations numerically under the normal SABR model against the market standards: Hagan's approximation and Monte Carlo simulation.

The structure of the paper is as follows. In Section 2, we will introduce the main concepts of the CMS market to make the paper self-contained. In Section 3, we explain how to derive a general expression for the convexity adjustment for CMS caps, CMS floors, and CMS swaps using Malliavin calculus. This expression will depend on the different quadratic payoffs explained above. In Section 4, we provide a brief explanation of the Watanabe calculus. We will also give an example of the normal SABR model contrasting the accuracy of the approximation. In Section 5, we show how to use Watanabe calculus to obtain the price of a quadratic call, quadratic put, and quadratic swaps under different models. The approximation is applied to local and stochastic local volatility models. In particular, the accuracy is assessed for the normal SABR model by comparing the results with those obtained from Monte Carlo simulations and Hagan's approximation. Finally, in Section 7, we will present our conclusions.

We provide the code to execute all the examples presented in the paper at doi: <https://github.com/Dagalon/PyStochasticVolatility>.

2. A brief introduction to the CMS market

One of the principal uses of quadratic payoffs is the convexity adjustment calculation, especially for CMS products. Therefore, to make the article self-contained, we do a brief introduction to the CMS market.

We will consider the market standard of two curves: one to discount the cash flows and another to estimate the floating coupon. These two curves are equivalent to the discount and repo curves in the equity market.

The paper will use the following notation:

- The discount curve will be denoted by d , and the estimated curve by e .
- $P^i(T)$ is the spot discount factor to T using the curve $i=d, e$.
- $P^i(t, T) = \frac{P^i(T)}{P^i(t)}$ is the forward discount factor associated to the curve $i = d, e$ from t to T .
- $\Delta(T_a, T_b)$ is the fraction's year between T_a and T_b .
- $L^i(t, T_a, T_b) = \frac{1}{\Delta(T_a, T_b)} \left(\frac{P^i(t, T_a)}{P^i(t, T_b)} - 1 \right)$ is the floating rate associated with the curve $i = d, e$.

The swap is the most commonly traded derivative in the interest rate market. While there are various swap types, the market standard is to exchange a fixed coupon for a floating one until the end of the contract. Fixed and floating rate payments can be scheduled on different dates and periods. But, to simplify the notation, we assume that payments are made on the same dates and in the same fractions of the year. Today is denoted as T_0 , and the future payment dates are scheduled by

$$\mathcal{T} = \{T_a, T_{a+1}, \dots, T_b\}.$$

The standard swap price is

$$\begin{aligned} V^{Swap}(T_0, T_a, T_b, C) \\ = w * \left(\sum_{i=a}^b \Delta(T_{i-1}, T_i) P^d(T_0, T_i) C \right. \\ \left. - \sum_{i=a}^b \Delta(T_{i-1}, T_i) P^d(T_0, T_i) \mathbb{E}_{T_0}^{T_i} [L^e(T_{i-1}, T_{i-1}, T_i)] \right). \end{aligned} \quad (5)$$

When we receive a fixed coupon, $w = 1$, we are in a receiver swap. On the other hand, if we pay the fixed coupon, $w = -1$, we are in a payer swap. The measure $\mathbb{Q}^{T_i}$ used on the floating leg is associated with the numeraire $N_t^{T_p} := P^d(t, T_p)$.

Within the scope of swaps, two essential concepts arise: the swap rate and the annuity factor. The swap rate denotes the fixed coupon that ensures the financial equilibrium of the swap structure, calculated using the following formula:

$$S_{a,b}(T_0) = \frac{\sum_{i=a}^b \Delta(T_{i-1}, T_i) P^d(T_0, T_i) \mathbb{E}_{T_0}^{T_i} [L^e(T_{i-1}, T_{i-1}, T_i)]}{\sum_{i=a}^b \Delta(T_{i-1}, T_i) P^d(T_0, T_i)}.$$

The annuity factor, likewise, is determined by the following formula:

$$A(T_0, T_a, T_b) = \sum_{i=a}^b \Delta(T_{i-1}, T_i) P^d(T_0, T_i).$$

A constant maturity swap (CMS) is a fixed-for-floating swap, where the floating leg periodically pays, or receives, a swap rate with a fixed time to maturity. It has the same structure

as the standard swap but changes the floating payment from $\mathbb{E}_{T_i}^T[L^e(T_{i-1}, T_{i-1}, T_i)]$ to $\mathbb{E}_{T_0}^{T_p}[S_{i-1,i-1+b}(T_{i-1})]$ with $T_{i-1} \leq T_{p_i} < T_i$.

The CMS swap price is given by

$$\begin{aligned} V_{CMS}^{Swap}(T_0, T_a, T_b, C) \\ = w * \left(\sum_{i=a}^b \Delta(T_{i-1}, T_i) P^d(T_0, T_i) C \right. \\ \left. - \sum_{i=a}^b \Delta(T_{i-1}, T_i) P^d(T_0, T_{p_i}) \mathbb{E}_{T_0}^{T_{p_i}} [S_{i-1,i-1+b}(T_{i-1})] \right). \end{aligned} \quad (6)$$

The valuation of standard and CMS swaps exhibits a significant difference. In the standard swap case, the floating coupon is a martingale under the forward measure $N_t^{T_p}$. Specifically,

$$\mathbb{E}_{T_0}^{T_i} [L^e(T_{i-1}, T_{i-1}, T_i)] = L^e(T_0, T_{i-1}, T_i).$$

By contrast, the swap rate does not satisfy this condition, leading to

$$\mathbb{E}_{T_0}^{T_p} [S_{i-1,i-1+b}(T_{i-1})] \neq S_{i-1,i-1+b}(T_0).$$

Instead, the swap rate is a martingale with respect to the measure induced by the numeraire $N_t^{a,b} := A(t, T_a, T_b)$. As the floating rate and the swap rate are martingales under different numeraries, a convexity adjustment is required for CMS products.

In fact, we have that

$$\begin{aligned} \mathbb{E}_{T_0}^{T_{p_i}} [S_{i-1,i-1+b}(T_{i-1})] \\ = \frac{N_{T_0}^{i-1,i-1+b}}{P^d(T_0, T_{p_i})} \mathbb{E}_{T_0}^{T_{p_i}} \left[S_{i-1,i-1+b}(T_{i-1}) \frac{N_{T_{i-1}}^{i-1,i-1+b}}{P^d(T_{i-1}, T_{p_i})} \right]. \end{aligned}$$

The essence of CMS modeling lies in defining a mapping function $M(t, a, b)$. A comprehensive discussion of the selection and requirements of this mapping function is presented in Andersen and Piterbarg (2010). In this work, we utilize the following mapping function:

$$M(t, a, b, p) = \frac{P^d(t, T_p)}{N_t^{a,b}}.$$

Then, we have that

$$\begin{aligned} \mathbb{E}_{T_0}^{T_{p_i}} [S_{i-1,i-1+b}(T_{i-1})] \\ = \mathbb{E}_{T_0}^{N^{i-1,i-1+b}} \left[S_{i-1,i-1+b}(T_{i-1}) \frac{M(T_{i-1}, i-1, i-1+b, p_i)}{M(T_0, i-1, i-1+b, p_i)} \right] \\ = S_{i-1,i-1+b}(T_0) + \mathbb{E}_{T_0}^{N^{i-1,i-1+b}} \left[S_{i-1,i-1+b}(T_{i-1}) \right. \\ \left. \left(\frac{M(T_{i-1}, i-1, i-1+b, p_i)}{M(T_0, i-1, i-1+b, p_i)} - 1 \right) \right]. \end{aligned} \quad (7)$$

The second term in the summand on the right side represents the convexity adjustment for the CMS rate; for further details, see Hagan (2003).

To conclude this section, we introduce CMS Caps and Floors. This paper aims to develop various price expansions over different models to calculate their convexity adjustments. A CMS Cap and CMS Floor consist of a series of call and put options, known as caplets or floorlets respectively, with the underlying $S_{a,b}(t)$. The payoff is given by:

$$\begin{aligned} V_{CMS}^{Cap}(T_0, T_a, T_b, K) &= \left(\sum_{i=a}^b \Delta(T_{i-1}, T_i) P^d(T_0, T_{p_i}) \right. \\ &\quad \left. \times \mathbb{E}_{T_0}^{T_{p_i}} [S_{i-1,i-1+b}(T_{i-1}) - K]^+ \right), \\ V_{CMS}^{Floor}(T_0, T_a, T_b, K) &= \left(\sum_{i=a}^b \Delta(T_{i-1}, T_i) P^d(T_0, T_{p_i}) \right. \\ &\quad \left. \times \mathbb{E}_{T_0}^{T_{p_i}} [K - S_{i-1,i-1+b}(T_{i-1})]^+ \right). \end{aligned}$$

Now, we focus on calculating the value of a single caplet, given by

$$\begin{aligned} P^d(0, T_p) \mathbb{E}^{T_p} [(S_{a,b}(T_a) - K)^+] \\ = P^d(0, T_a) \mathbb{E}^{T_a} [(S_{a,b}(T_a) - K)^+ P^d(T_a, T_p)]. \end{aligned}$$

Using a change of numeraire, we have that

$$\begin{aligned} N_0^{a,b} \mathbb{E}^{N^{a,b}} [(S_{a,b}(T_a) - K)^+ M(T_a, a, b, p)] \\ = N_0^{a,b} \mathbb{E}^{N^{a,b}} [(S_{a,b}(T_a) - K)^+] \\ + N_0^{a,b} \mathbb{E}^{N^{a,b}} [(S_{a,b}(T_a) - K)^+ (M(T_a, a, b, p) - 1)] \end{aligned}$$

where $N_0^{a,b} = A(0, a, b)$ and $M(t, a, b, p) = \frac{P(t, T_p)}{A(t, a, b)}$ correspond to the convexity adjustment for a CMS Cap. To calculate this term, it is generally assumed that

$$M(t, a, b, p) = M_p(S_{a,b}(t)).$$

Assuming there is a payoff function $g \in \mathcal{C}^2$, then we apply the following replication formula

$$I(S_{a,b}(t) > K) g(S_{a,b}(t)) = \int_K^\infty (S_{a,b}(t) - u)^+ g''(u) du$$

with

$$g(S_{a,b}(t)) = (S_{a,b}(t) - K) (M_p(S_{a,b}(t)) - 1).$$

However, a primary concern with this formula is that it requires knowledge of the swaption price along the strike grid, as the right-hand side tends to ∞ . Consequently, the convexity adjustment is highly sensitive to the extrapolation assumptions applied in practice.

In recent papers by Hagan *et al.* (2019) and Hagan and Woodward (2021), the authors propose a new methodology that expresses the convexity adjustment for CMS Caps, CMS

Floors, and CMS Swaps as a quadratic payoff. By employing the singular perturbation technique, they obtain an expansion for the implied volatility of the quadratic payoff under the dynamics of the SABR-type model.

In the following sections, we will see how to use Malliavin calculus to express the convexity adjustment of a CMS Cap, CMS Floor, and CMS Swap as a function of a quadratic payoff. Next, the following sections will illustrate how to use Watanabe calculus to compute the price of a quadratic payoff within a general stochastic local volatility model.

3. The connection between quadratic payoffs and the CMS cap-floor convexity

In this section, we illustrate the connection between quadratic payoffs and the convexity adjustment for CMS Cap, Floor, and Swap instruments. We employ Malliavin's techniques to derive a general representation for this adjustment. For more detailed content on Malliavin calculus, refer to Nualart (1995) and Alòs and García Lorite (2021).

We will start with the convexity adjustment for a CMS Cap, as detailed in Section 2. We have

$$\begin{aligned} CA_{CMS}^{cap}(0, a, b) &= N_0^{a,b} \mathbb{E}^{N^{a,b}} \left[(S_{a,b}(T_a) - K)^+ \right. \\ &\quad \times (M(T_a, a, b, p) - 1) \Big] \\ &= N_0^{a,b} \mathbb{E}^{N^{a,b}} \left[(S_{a,b}(T_a) - K)^+ \right. \\ &\quad \times \left(\mathbb{E}^{N^{a,b}} [M(T_a, a, b, p) | S_{a,b}(T_a)] - 1 \right) \Big]. \end{aligned}$$

Here, we define the mapping function $M(S_{a,b}(T_a))$ as

$$M(S_{a,b}(T_a)) := \mathbb{E}^{N^{a,b}} [M(T_a, a, b, p) | S_{a,b}(T_a)].$$

Thus,

$$\begin{aligned} CA_{CMS}^{cap}(0, a, b) &= N_0^{a,b} \mathbb{E}^{N^{a,b}} \left[(S_{a,b}(T_a) - K)^+ (M(S_{a,b}(T_a)) - 1) \right]. \quad (8) \end{aligned}$$

The convexity adjustment depends on the choice of the mapping function. A widely employed choice is the linear model, where it is assumed that

$$M(S_{a,b}(T_a)) = \alpha(T_a) S_{a,b}(T_a) + \beta(T_a).$$

The parameters $\alpha(T_a)$ and $\beta(T_a)$ are chosen based on the properties of $S_{a,b}(T_a)$ under the measure associated with the numeraire $N_t^{a,b}$.

In the following proposition, we establish a connection between the convexity adjustment for a CMS Cap and quadratic payoffs.

PROPOSITION 3.1 Assume that the process $S_{a,b}(t)$ is in $\mathbb{D}^{1,2}(L^2[0, T])$, where $\mathbb{D}^{1,2}(L^2[0, T])$ is the process space in $L^2(\Omega, [0, T])$ with Malliavin derivatives in $L^2(\Omega, [0, T])$.

Then, the convexity adjustment for a CMS Cap can be approximated as follows:

$$\begin{aligned} CA_{CMS}^{cap}(0, a, b) &\approx \left[(M(S_{a,b}(0)) - 1) + \partial_S M(S_{a,b}(0)) \right. \\ &\quad \times (K - S_{a,b}(0)) \Big] V^C(T_a, S_{a,b}, K) \\ &\quad + \partial_S M(S_{a,b}(0)) V^{QC}(T_a, S_{a,b}, K). \end{aligned}$$

Proof Since $S_{a,b}(t)$ is an L^2 -valued stochastic process, the Clark-Ocone formula can be applied to $F_t = S_{a,b}(t)$. Thus, we have:

$$\begin{aligned} M_{S_{a,b}(T_a)} &= M_{S_{a,b}(0)} + \int_0^{T_a} \mathbb{E}_s^{N^{a,b}} [\partial_S M(S_{a,b}(T_a))] \\ &\quad \times D_s^W S_{a,b}(T_a) dW_s. \end{aligned}$$

Freezing $\partial_S M(S_{a,b}(T_a))$ at $S_{a,b}(0)$, we obtain the approximation:

$$\begin{aligned} M_{S_{a,b}(T_a)} &\approx M_{S_{a,b}(0)} + \partial_S M(S_{a,b}(0)) \\ &\quad \times \int_0^{T_a} \mathbb{E}_s^{N^{a,b}} [D_s^W S_{a,b}(T_a)] dW_s \\ &= M(S_{a,b}(0)) + \partial_S M(S_{a,b}(0)) (S_{a,b}(T_a) - S_{a,b}(0)). \end{aligned}$$

Substituting this approximation into (8) concludes the proof. $\blacksquare$

REMARK 1 A similar approach can be applied to the CMS Floor case, leading to:

$$\begin{aligned} CA_{CMS}^{floor}(0, a, b) &\approx \left[(M(S_{a,b}(0)) - 1) + \partial_S M(S_{a,b}(0)) \right. \\ &\quad \times (K - S_{a,b}(0)) \Big] V^P(T_a, S_{a,b}, K) \\ &\quad - \partial_S M(S_{a,b}(0)) V^{QP}(T_a, S_{a,b}, K). \end{aligned}$$

For the CMS Swap, we find:

$$\begin{aligned} CA_{CMS}^{swap}(0, a, b) &\approx \left[(M(S_{a,b}(0)) - 1) + \partial_S M(S_{a,b}(0)) \right. \\ &\quad \times (K - S_{a,b}(0)) \Big] (S_{a,b}(0) - K) \\ &\quad + \partial_S M(S_{a,b}(0)) V^{QS}(T_a, S_{a,b}, K). \end{aligned}$$

4. Basic introduction to Watanabe's expansion

Malliavin calculus has become an essential tool for justifying asymptotic expansions of option payoffs. A notable contribution is Theorem 2.3 from Watanabe (1987), which provides the foundational framework for constructing these expansions. In particular, Kunitomo and Takahashi (2003) introduced a methodology for pricing financial contingent claims, where the underlying asset prices are modeled by a general class of continuous Itô processes. This technique is known as Small Disturbance Asymptotics or Watanabe's expansion.

The rigorous basis of this approach lies in the infinite-dimensional framework of the Malliavin-Watanabe calculus, originally developed by Watanabe (1987) as a probabilistic

theory. The theory was further refined by Yoshida (1992a) and Yoshida (1992b), broadening its applications to statistical inference.

Additional works, such as Osajima (2006), Kahl (2008), Takahashi *et al.* (2012), Takahashi and Yamada (2012), and Muroi (2021), further illustrate the practical use of this methodology in quantitative finance.

Let $\mathbb{D}^\infty$ denote the space of smooth Wiener functionals. Consider an underlying process F_t governed by an SDE with a unique strong solution. As shown in Theorem 3 of Kunitomo and Takahashi (2003), F_t admits an asymptotic expansion in terms of a small parameter $\epsilon \ll 1$. Thus, $F_\epsilon(T)$ has the following asymptotic expansion in $\mathbb{D}^\infty$:

$$\begin{aligned} F_\epsilon(T) &= F_0 + \epsilon \cdot g_1(T) + \epsilon^2 \cdot g_2(T) + \epsilon^3 \cdot g_3(T) \\ &\quad + \dots + \epsilon^n \cdot g_n(T) + \mathcal{O}(\epsilon^{n+1}) \\ &= F_0 + \epsilon \sigma_0 \sqrt{T} (\hat{g}_1(T) + \epsilon \hat{g}_2(T) + \epsilon^2 \hat{g}_3(T) \\ &\quad + \dots + \epsilon^{n-1} \cdot \hat{g}_n(T)) + \mathcal{O}(\epsilon^{n+1}) \end{aligned} \quad (9)$$

where

$$\hat{g}_i(T) = \frac{g_i(T)}{\sigma_0 \sqrt{T}}.$$

The expansion is constructed in such a way that $\hat{g}_1 \sim \mathcal{N}(0, 1)$. Here, σ_0 denotes the initial volatility and T as the time to maturity. The auxiliary term ϵ is introduced to derive the asymptotic option pricing formula.

A crucial condition for obtaining these expansions is the non-degeneracy of the Malliavin covariance matrix γ_{F_ϵ} , which guarantees that the process is sufficiently regular. For any $n \geq 1$, this non-degeneracy condition is expressed as:

$$\limsup_{\epsilon \rightarrow 0} \left\| \frac{1}{\det(\gamma_{F_\epsilon})} \right\|_p < \infty,$$

ensuring that the expansion remains valid as $\epsilon \rightarrow 0$.

Yoshida extended Watanabe's result by relaxing the regularity conditions, ensuring that even if the payoff function lacks smoothness, the asymptotic expansion remains valid under a localized non-degeneracy condition. Furthermore, Kunitomo and Takahashi (2003) showed that these conditions can be lowered when the underlying process is governed by an SDE with a unique strong solution.

Once we have ensured the existence of the expansion (9) for the underlying $F_\epsilon(T)$, it is possible to obtain an approximation for

$$V^C(T, F_T, K) := \mathbb{E}^{\mathbb{Q}^N} [(F_T - K)^+].$$

Now, if we substitute (9) in the above expression, we get that

$$\begin{aligned} &\frac{V^C(T, F, K)}{N_T} \\ &= \mathbb{E}^{\mathbb{Q}^N} \left[\left(x_0 - K + \epsilon \sigma_0 \sqrt{T} \sum_{j=1}^{\infty} \epsilon^{j-1} \hat{g}_j(T) \right)^+ \right] \end{aligned}$$

$$= \epsilon \sigma_0 \sqrt{T} \mathbb{E}^{\mathbb{Q}^N} \left[\left(\hat{g}_1(T) - y + \epsilon \sum_{j=1}^{\infty} \epsilon^{j-1} \hat{g}_{j+1}(T) \right)^+ \right] \quad (10)$$

where

$$y = \frac{K - F_0}{\epsilon \sigma_0 \sqrt{T}}. \quad (11)$$

Furthermore, we define

$$V_{(j)}^C := \partial_F^{(j)} (F - K)^+|_{F=\hat{g}_1(T), K=y}.$$

By applying Taylor's expansion to $V^C(T, F, K)$ around $\hat{g}_1(T) - y$ and considering all the derivatives of $(F - K)$ in the sense of distributions, we derive the following price expansion:

$$\begin{aligned} &\frac{V^C(T, F, K)}{N_T} \\ &= \epsilon \sigma_0 \sqrt{T} \mathbb{E}^{\mathbb{Q}^N} \left[(\hat{g}_1(T) - y)^+ + \epsilon V_{(1)}^C \hat{g}_2(T) \right. \\ &\quad + \epsilon^2 \left(V_{(1)}^C \hat{g}_3(T) + \frac{1}{2} V_{(2)}^C \hat{g}_2^2(T) \right) \\ &\quad + \epsilon^3 \left(V_{(1)}^C \hat{g}_4(T) + V_{(2)}^C \hat{g}_2(T) \hat{g}_3(T) + \frac{1}{6} V_{(3)}^C \hat{g}_2^3(T) \right) \\ &\quad + \epsilon^4 \left(V_{(1)}^C \hat{g}_5(T) + V_{(2)}^C \hat{g}_2(T) \hat{g}_4(T) + \frac{1}{2} \hat{g}_3^2(T) \right) \\ &\quad \left. + \frac{1}{2} V_{(3)}^C \hat{g}_2^2(T) \hat{g}_3(T) + \frac{1}{24} V_{(4)}^C \hat{g}_2^2(T) \right) \Big] + \mathcal{O}(\epsilon^6). \end{aligned} \quad (12)$$

Now, if we use the tower rule by conditioning on $W_T = \hat{g}_1(T) \sqrt{T}$, we have that

$$\begin{aligned} &\frac{V^C(T, F, K)}{N_T} \\ &= \epsilon \sigma_0 \sqrt{T} \mathbb{E}^{\mathbb{Q}^N} \left[(\hat{g}_1(T) - y)^+ + \epsilon V_{(1)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2(T) | W_T] \right. \\ &\quad + \epsilon^2 \left(V_{(1)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_3(T) | W_T] + \frac{1}{2} V_{(2)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2^2(T) | W_T] \right) \\ &\quad + \epsilon^3 \left(V_{(1)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_4(T) | W_T] + V_{(2)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2(T) \hat{g}_3(T) | W_T] \right. \\ &\quad \left. + \frac{1}{6} V_{(3)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2^3(T) | W_T] \right) \\ &\quad + \epsilon^4 \left(V_{(1)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_5(T) | W_T] + V_{(2)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2(T) \hat{g}_4(T) | W_T] \right. \\ &\quad + \frac{1}{2} \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_3^2(T) | W_T] + \frac{1}{2} V_{(3)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2^2(T) \hat{g}_3(T) | W_T] \\ &\quad \left. \left. + \frac{1}{24} V_{(4)}^C \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2^2(T) | W_T] \right) \right] + \mathcal{O}(\epsilon^6). \end{aligned} \quad (13)$$

EXAMPLE 4.1 This example illustrates the application of Watanabe calculus for approximating the price of a call option

when the underlying asset follows a normal SABR model. We consider the forward price dynamics described by:

$$\begin{aligned} dF_t &= \sigma_t dW_t \\ d\sigma_t &= \nu \sigma_t dB_t. \end{aligned}$$

where $\langle dW_t, dB_t \rangle = \rho dt$. Assuming that σ_t is small, we introduce a parameter ϵ such that $\sigma_t = \epsilon \sigma_{\epsilon,t}$. Consequently, the forward price dynamics can be expressed as:

$$\begin{aligned} dF_{\epsilon,t} &= \epsilon \sigma_{\epsilon,t} dW_t \\ d\sigma_{\epsilon,t} &= \nu \sigma_{\epsilon,t} dB_t. \end{aligned} \quad (14)$$

LEMMA 4.2 *Under the dynamics specified by (14), the forward price can be rewritten as:*

$$\begin{aligned} F_{\epsilon,T} &= F_0 + \epsilon \sigma_0 W_T + \nu \epsilon^2 \sigma_0 \int_0^T \int_0^u dB_{u_1} dW_u \\ &\quad + \nu^2 \epsilon^3 \sigma_0 \int_0^T \int_0^u \int_0^{u_1} dB_{u_2} dB_{u_1} dW_u + \dots \\ &\quad + \nu^{n-1} \epsilon^n \sigma_0 \int_0^T \int_0^u \int_0^{u_1} \dots \int_0^{u_{n-2}} dB_{u_{n-1}} \dots dB_{u_1} dW_u \\ &\quad + \nu^n \epsilon^{n+1} \int_0^T \int_0^u \int_0^{u_1} \dots \int_0^{u_{n-1}} \sigma_{\epsilon,u_n} dB_{u_n} \dots dB_{u_1} dW_u. \end{aligned}$$

Proof The result follows directly from the iterative application of the Itô-Taylor expansion. ■

For practical purposes, we aim to derive an approximation for the option price. Thus, we consider the fourth-order forward price dynamics:

REMARK 2 The 4th order approximation of the forward price dynamics is

$$\begin{aligned} F_{\epsilon,T} &= F_0 + \epsilon \sigma_0 W_T + \epsilon^2 \nu \sigma_0 \int_0^T \int_0^u dB_{u_1} dW_u \\ &\quad + \epsilon^3 \nu^2 \sigma_0 \int_0^T \int_0^u \int_0^{u_1} dB_{u_2} dB_{u_1} dW_u \\ &\quad + \epsilon^4 \nu^3 \sigma_0 \int_0^T \int_0^u \int_0^{u_1} \int_0^{u_2} dB_{u_3} dB_{u_2} dB_{u_1} dW_u + \mathcal{O}(\epsilon^4). \end{aligned}$$

To obtain the Watanabe price expansion (12), we need to find an expression for the $\hat{g}_i$ terms. Using Remark 2, we can straightforwardly derive the analytical expressions for each term as follows:

$$\hat{g}_1(T) = \frac{W_T}{\sqrt{T}},$$

$$\hat{g}_2(T) = \frac{\nu}{\sqrt{T}} \int_0^T \int_0^u dB_{u_1} dW_u,$$

$$\hat{g}_3(T) = \frac{\nu^2}{\sqrt{T}} \int_0^T \int_0^u \int_0^{u_1} dB_{u_2} dB_{u_1} dW_u$$

$$\hat{g}_4(T) = \frac{\nu^3}{\sqrt{T}} \int_0^T \int_0^u \int_0^{u_1} \int_0^{u_2} dB_{u_3} dB_{u_2} dB_{u_1} dW_u. \quad (15)$$

Additionally, we have the following identities for the option price derivatives:

$$V_{(1)}^C(F, K) = I(F > K), \quad (16)$$

$$V_{(2)}^C(F, K) = \delta(F - K), \quad (17)$$

$$V_{(3)}^C(F, K) = \partial_F \delta(F - K), \quad (18)$$

where I denotes the indicator function, δ represents the Dirac delta function and

$$\int_{\mathbb{R}} f(u) \partial_F \delta(u - K) du = -f'(K) \quad \forall f \in \mathcal{C}_0^\infty(\mathbb{R}).$$

Watanabe's expansion is based on the computation of conditional expectations of iterated integrals, which can be transformed into unconditional expectations using Hermite polynomials, as detailed in Takahashi *et al.* (2012). An efficient algorithm for computing these conditional expectations is discussed in, for instance, Li (2013), Li (2014), and Li *et al.* (2016). This algorithm extends previous approaches by employing combinatorial analysis to derive a novel closed-form expression for the conditional expectation of products of iterated Itô integrals.

We now present an option price expansion under the Normal SABR model, which was previously proved in Takahashi *et al.* (2011), and later refined in Takahashi *et al.* (2012). The objective of the theorem and its proof is to illustrate the application of the Watanabe expansion. This forms the basis for its extension to quadratic payoffs in the following sections.

THEOREM 4.3 ($T^{\frac{3}{2}}$ order option Price Expansion under the Normal SABR Model) *Consider the forward price dynamics governed by the normal SABR model as specified in (14). The 2nd-order call option approximation price is given by*

$$\begin{aligned} &\frac{V^C(T, F, K)}{N_T} \\ &= \sigma_0 \sqrt{T} \left(G(y) + \frac{1}{2} \rho \nu y \phi(y) \sqrt{T} \right. \\ &\quad \left. + \nu^2 \left(\frac{(4\rho^2(y^2 - 1) + 3\rho^2(y^2 - 1)^2 + 2\bar{\rho}^2(2y^2 + 1))}{24} \right) \phi(y) T \right. \\ &\quad \left. + \frac{\nu^3 \rho y (\rho^2 y^6 - 9\rho^2 y^4 + 15\rho^2 y^2 - 7\rho^2 + 4\bar{\rho}^2 y^4 - 18\bar{\rho}^2 y^2 + 2\bar{\rho}^2)}{48} \phi(y) T^{\frac{3}{2}} \right) \end{aligned}$$

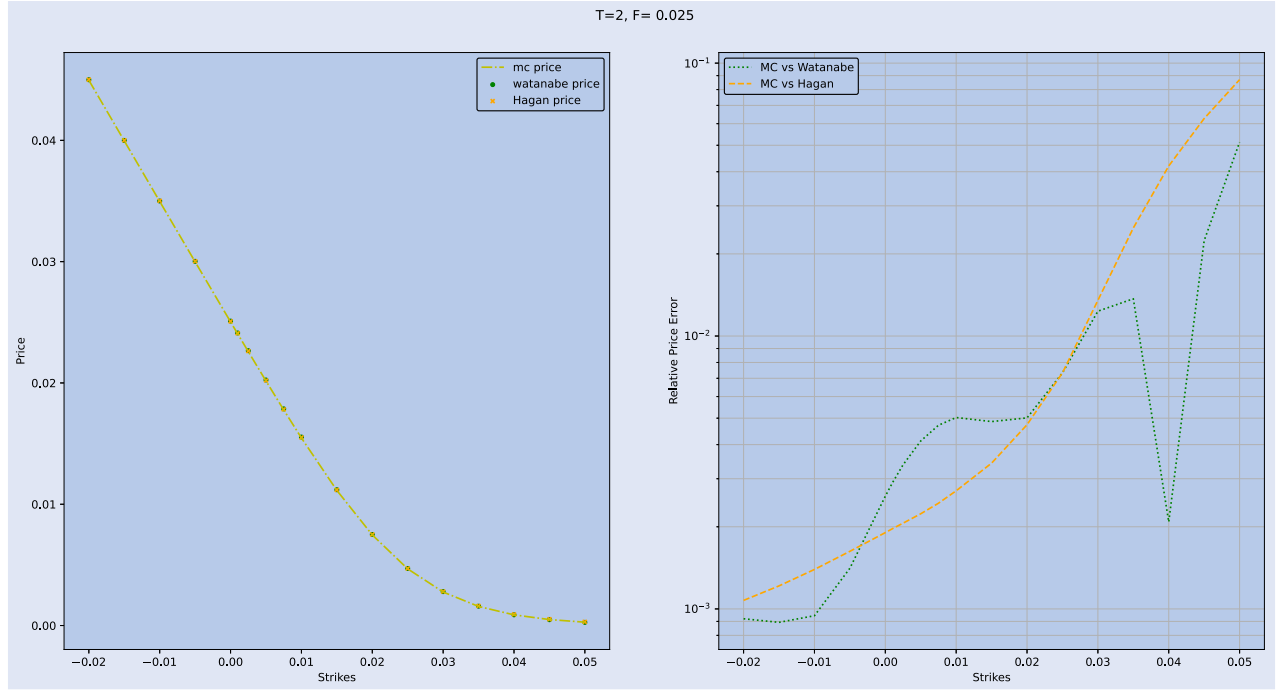


Figure 1. Comparison of call options using Watanabe's expansion against the industry standards. The parameters considered are $\alpha = 0.008188$, $\nu = 0.3620$, $\rho = 0.3137$, and $T = 2Y$.

$$+ \frac{\nu^3 \rho^3 y (y^2 - 3) (4y^2 - 3)}{24} \phi(y) T^{\frac{3}{2}} \Bigg) + \mathcal{O}(T^{\frac{5}{2}}), \quad (19)$$

where y is defined as in (11), $G(y) = \phi(y) - y\bar{\Phi}(y)$, $\phi(\cdot)$ denotes the standard normal pdf, $\Phi(\cdot)$ represents the standard normal cdf, and $\bar{\Phi}(\cdot) = 1 - \Phi(\cdot)$ is the complementary cdf.

Proof To derive the price expansion, we consider the price expansion provided in (13) up to order ϵ^4 . Next, we compute the terms $\hat{g}_i$ as detailed in Appendix 1. Finally, if we take the limit $\epsilon \rightarrow 1$, we arrive at the desired price approximation. In addition, the error is bounded as $|E(T)| \leq CT^2$ through the Malliavin's integration by parts (see more details in Takahashi and Yamada 2016). ■

REMARK 3 Although Hagan's and Watanabe's approximations are derived and expressed differently, Hagan's approximation is formulated in terms of implied volatility, while Watanabe's expansion is applied directly to option prices, a quantitative comparison can be made at the at-the-money (ATM) point, i.e. when $K = F$. The Watanabe approximation simplifies to:

$$\frac{V^C(T, F, K)}{N_T} = \sigma_0 \sqrt{T} \left(1 + \frac{2\nu^2 - 3\nu^2 \rho^2}{24} \frac{T}{\sqrt{2\pi}} \right) + \mathcal{O}(T^{\frac{5}{2}}).$$

Meanwhile, the Bachelier at-the-money implied volatility is:

$$\sigma_B(T, F) = \sigma_0 \left(1 + \frac{2\nu^2 - 3\nu^2 \rho^2}{24} T \right) + \mathcal{O}(T^2),$$

Table 1. Normal SABR parameters considering an underlying swap tenor of 5 years.

SABR Parameters	2Y	5Y	10Y
α	0.008188	0.007645	0.007183
ν	0.3620	0.3002	0.1969
ρ	0.3137	0.3421	0.3847

Thus, we recover the at-the-money implied volatility derived in Hagan *et al.* (2002). As the reader can observe, the two formulas are equivalent at the ATM, confirming the accuracy of the Watanabe approximation.

To determine the accuracy of the approximation in (19), we employ a Monte Carlo method alongside the implied volatility approximation proposed by Hagan *et al.* (2002) across a range of strike prices. The parameters presented in table 1 are calibrated to reflect the swaption market, specifically for an underlying swap tenor of 2 years, with maturities of 2, 5, and 10 years. These maturities are chosen due to their higher liquidity in the swaptions market with a 2-year underlying swap tenor.

Figures 1–3 compare the performance of the Watanabe approximation, Hagan's approximation, and Monte Carlo simulation. The analysis considers both absolute price differences and relative price errors. The relative error, with Monte Carlo prices as the benchmark, measures the normalized deviation of the approximated prices from the Monte Carlo results. In other words, the relative error for Hagan's approximation is defined as $\left| \frac{\text{Price Hagan}}{\text{Price MC}} - 1 \right|$. The Monte Carlo simulations were conducted with 300,000 paths and 50 time steps per year, employing antithetic sampling and variance matching techniques for improved accuracy.

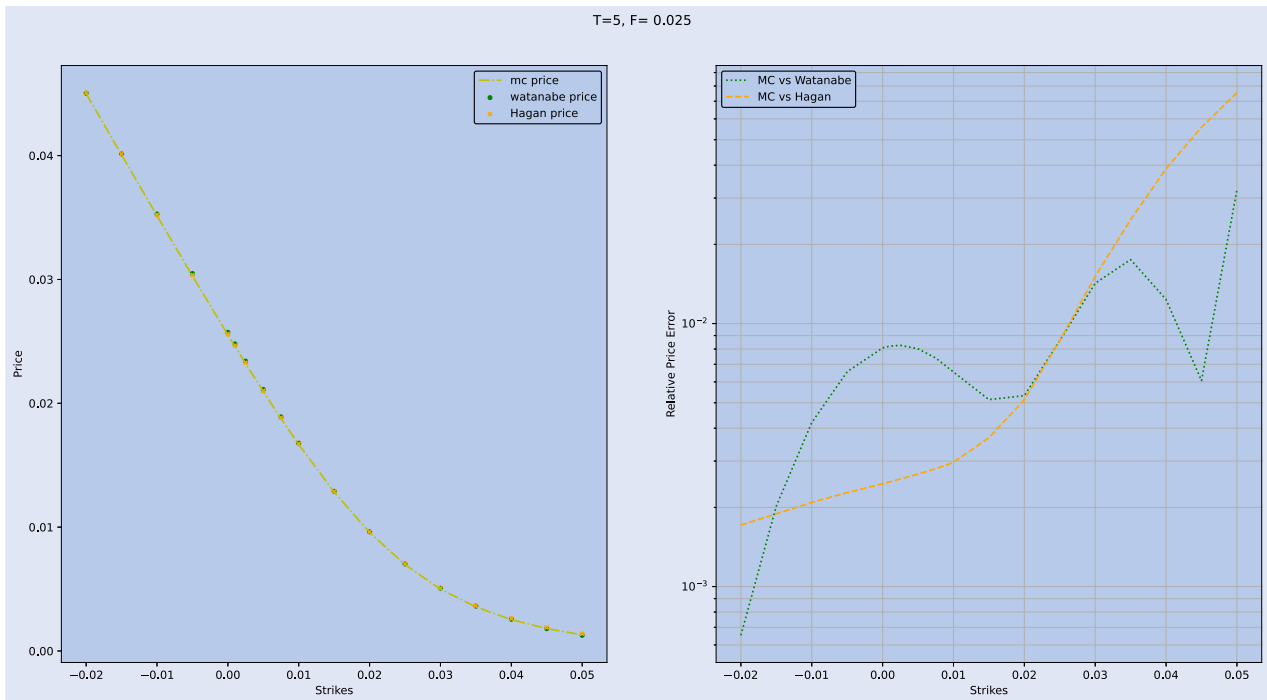


Figure 2. Comparison of call options using Watanabe's expansion against the industry standards. The parameters considered are $\alpha = 0.007645$, $\nu = 0.3002$, $\rho = 0.3421$, and $T = 5Y$.

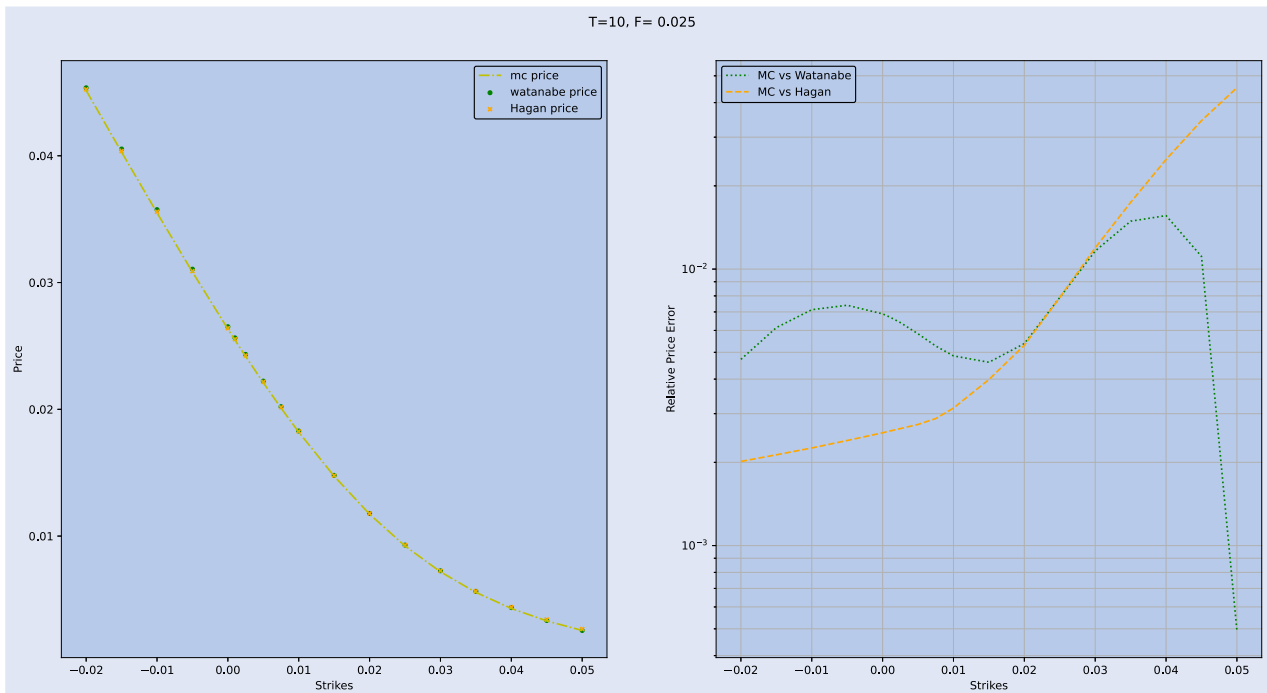


Figure 3. Comparison of call options using Watanabe's expansion against the industry standards. The parameters considered are $\alpha = 0.007183$, $\nu = 0.1969$, $\rho = 0.3847$, and $T = 10Y$.

In the following figures, we observe the numerical performance of the approximation.

In figures 1–3, we observe that the price comparisons across the three methodologies are consistent. We use the Monte Carlo method as a benchmark, with an estimated simulated error of 5×10^{-5} approximately, based on the standard deviation of the simulations.

All three figures illustrate that Hagan's and Watanabe's approximations perform comparably, as both methods exhibit

errors of a similar magnitude. Hagan's approximation shows greater accuracy on the left side of the tail; however, this accuracy decreases as we move to the right side. A similar trend is observed for Watanabe's approximation, except for the 10-year case.

In summary, both approximations are equally reliable, with Watanabe's approximation proving as effective as Hagan's, one of the industry standards. While it may not immediately appear superior, its remarkable flexibility enables it to

accommodate a diverse range of payoffs and models, consistently delivering the same level of accuracy as the top industry benchmarks. This adaptability makes it a valuable tool for practitioners.

5. Quadratic option price

In this section, following the methodology presented in Section 4, we derive a Watanabe expansion for quadratic payoff derivatives under three different dynamics: stochastic volatility, local volatility, and local stochastic volatility models. It should be noted that the quadratic payoff $V^{QC}(F, K)$ is more influenced by the tail of the underlying distribution than the price of the call option $V^C(T, F, K)$. This follows directly from formula (4), where we showed that $V^{QC}(F, K)$ can be replicated as follows:

$$(F_T - K)_+^2 = 2 \int_K^\infty (F_T - K')_+ dK'.$$

From the above equation, we observe the dependence of $V^{QC}(F, K)$ on the volatility smile when the strike is above K .

As discussed in Section 4, we find the Watanabe price expansion for the quadratic call, put, and swap.

THEOREM 5.1 (Price Approximation for Quadratic Calls, Puts, and Swaps) *Assume that there exists a forward price expansion as described by (9). Let $y(k)$ be defined by (11), $I(\cdot)$ be defined in (16) and $\phi(y)$ is the density function of a standard normal. Then, the $\mathcal{O}(\epsilon^4)$ -approximation for Watanabe's expansion of a quadratic call option price is*

$$\begin{aligned} \frac{V^{QC}(T, F, K)}{N_T} &= \epsilon^2 \sigma^2 (F_0) T \mathbb{E}^{\mathbb{Q}^N} \left[((\hat{g}_1(T) - y)^+)^2 2\epsilon (\hat{g}_1(T) - y)^+ \hat{g}_2(T) \right. \\ &\quad + \epsilon^2 \left(2(\hat{g}_1(T) - y)^+ \hat{g}_3(T) + I(\hat{g}_1(T) > y) \right. \\ &\quad \times \hat{g}_2^2(T)) + e^3 \left(2(\hat{g}_1(T) - y)^+ \hat{g}_4(T) + 2I(\hat{g}_1(T) > y) \right. \\ &\quad \times \hat{g}_2(T) \hat{g}_3(T) + \frac{\delta(\hat{g}_1(T) - y)}{3} \hat{g}_2^3(T) \left. \right) \left. \right] \\ &\quad + \mathcal{O}(\epsilon^6). \end{aligned} \quad (20)$$

The $\mathcal{O}(\epsilon^4)$ -approximation of a quadratic put option price is

$$\begin{aligned} \frac{V^{QP}(T, F, K)}{N_T} &= \epsilon^2 \sigma^2 (F_0) T \mathbb{E}^{\mathbb{Q}^N} \left[((y - \hat{g}_1(T))^+)^2 \right. \\ &\quad - 2\epsilon (y - \hat{g}_1(T))^+ \hat{g}_2(T) \\ &\quad + \epsilon^2 \left(I(y > \hat{g}_1(T)) \hat{g}_2^2(T) - 2(y - \hat{g}_1(T))^+ \hat{g}_3(T) \right) \\ &\quad \left. + e^3 (-2(y - \hat{g}_1(T))^+ \hat{g}_4(T) \right. \end{aligned}$$

$$\left. + 2I(y > \hat{g}_1(T)) \hat{g}_2(T) \hat{g}_3(T) - \frac{\delta(\hat{g}_1(T) - y)}{3} \hat{g}_2^3(T) \right) \left. \right] + \mathcal{O}(\epsilon^6). \quad (21)$$

The $\mathcal{O}(\epsilon^4)$ -approximation of a quadratic swap price is

$$\begin{aligned} \frac{V^{QS}(T, F, K)}{N_T} &= \epsilon^2 \sigma^2 (F_0) T \mathbb{E}^{\mathbb{Q}^N} \left[2\epsilon ((\hat{g}_1(T) - y))^2 + 2\epsilon^2 \hat{g}_2^2(T) \right. \\ &\quad \left. + 2\epsilon^3 (\hat{g}_1(T) - y)(\hat{g}_3(T) + \hat{g}_4(T) + \hat{g}_2(T) \hat{g}_3(T)) \right] \\ &\quad + \mathcal{O}(\epsilon^6). \end{aligned} \quad (22)$$

Proof We utilize the approach presented in (12). Using the forward expansion (9), we rewrite the option price. Following this, we apply a Taylor expansion around $\hat{g}_1(T) - y$ to $V^i(T, F, K)$ with $i = QC, QP, QS$. ■

5.1. Quadratic option under local volatility

When the volatility is constant, i.e. $\sigma(\cdot) = \sigma$, the price of the quadratic call option is given by

$$\begin{aligned} \frac{V^{QC}(T, F, K)}{N_T} &= ((F_0 - K)^2 + \sigma^2 T) \Phi\left(\frac{K - F_0}{\sigma\sqrt{T}}\right) \\ &\quad + (F_0 - K) \sigma \sqrt{T} \phi\left(\frac{K - F_0}{\sigma\sqrt{T}}\right). \end{aligned} \quad (23)$$

Here, $\Phi(\cdot)$ and $\phi(\cdot)$ denote the cumulative distribution function and the probability density function, respectively, for the standard normal distribution. We aim to construct an option price expansion based on (23), when the dynamic of the underlying is:

$$dF_{\epsilon,t} = \epsilon \sigma(F_{\epsilon,t}) dW_t^N \quad (24)$$

where $\sigma(\cdot)$ is a positive function that belongs to $\mathcal{C}_b^n(\mathbb{R})$ with $n \geq 4$.

LEMMA 5.2 *Under the dynamics described by (24), the forward price dynamics can be rewritten as follows:*

$$F_{\epsilon,T} = F_0 + \sum_{k=1}^{n-1} \int_0^T \frac{\partial^{k-1} \sigma(F_{u,\epsilon})|_{\epsilon=0}}{(k-1)!} dW_u^{\epsilon^k} + E(T, \epsilon), \quad (25)$$

where $E(T, \epsilon) = \int_0^\epsilon \int_0^{\epsilon_1} \int_0^{\epsilon_2} \dots \int_0^{\epsilon_{n-1}} \partial_\epsilon^n \sigma(F_{\epsilon_n, T}) dW_{\epsilon_n}^N dW_{\epsilon_{n-1}}^N \dots dW_{\epsilon_1}^N$ and $\sigma(\cdot) \in \mathcal{C}_b^n(\mathbb{R})$.

Proof If we take iteratively ∂_ϵ in (24), we can show that

$$\partial_\epsilon^n F_{\epsilon,T} = n \int_0^T \partial_\epsilon^{n-1} F_{\epsilon,u} dW_u + \epsilon \int_0^T \partial_\epsilon^n F_{\epsilon,u} dW_u$$

Now, if we apply Taylor expansion around $\epsilon = 0$ we get that

$$F_{\epsilon,T} = F_0 + \sum_{k=1}^{n-1} \frac{\partial_\epsilon^k F_{\epsilon,T}|_{\epsilon=0}}{k!} \epsilon^k + \int_0^\epsilon \int_0^{\epsilon_1} \int_0^{\epsilon_2}$$

$$\dots \int_0^{\epsilon_{n-1}} \partial_{\epsilon}^n \sigma(F_{\epsilon_n, T}) d\epsilon_n d\epsilon_{n-1} \dots d\epsilon_1.$$

■

As mentioned earlier, obtaining the Watanabe price expansion requires deriving an expression for the $\hat{g}_i$ terms. Comparing Lemma 5.2 with the price expansion (9), we have that

$$\begin{aligned} \hat{g}_1(T) &= \frac{W_T}{\sqrt{T}} \\ \hat{g}_2(T) &= \partial_F \sigma(F_0) \frac{I_{(1,1)}(T)}{\sqrt{T}} \\ \hat{g}_3(T) &= \frac{\sigma(F_0) \partial_F^2 \sigma(F_0)}{2\sqrt{T}} \left(\frac{W_T^2}{3} - \int_0^T W_u du \right) \\ &\quad + \frac{(\partial_F \sigma(F_0))^2}{\sqrt{T}} I_{(1,1,1)}(T) \\ \hat{g}_4(T) &= \frac{\partial_F^3 \sigma(F_0) \sigma^2(F_0)}{24\sqrt{T}} \left(\int_0^T \frac{W_u^4}{4} - \frac{3}{2} \int_0^T W_u^2 dW_u \right) \\ &\quad + \frac{\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) C(F_0)}{8\sqrt{T}} \int_0^T (W_u^3 - W_u) dW_u \\ &\quad + \frac{\partial_F \sigma(F_0)}{24} \int_0^T \hat{g}_3(u) dW_u. \end{aligned} \quad (26)$$

where $I_{(1,\dots,1)}(T) = \int_0^T \int_0^{u_1} \dots \int_0^{u_{n-1}} dW_n dW_{n-1} \dots dW_1$.

THEOREM 5.3 (Local volatility expansion for quadratic payoffs) *Under the local volatility model (24), consider y is given by (11), $\phi(\cdot)$ is standard normal pdf, $\Phi(\cdot)$ the standard normal cdf and $\bar{\Phi}(\cdot) = 1 - \Phi(\cdot)$. Then, the $T^{\frac{3}{2}}$ -order approximation for the price of a quadratic call option is*

$$\begin{aligned} \frac{V^{QC}(T, F, K)}{N_T} &= \sigma^2(F_0)T \left(G_q^c(y) + \partial_F \sigma(F_0) \phi(y) \sqrt{T} \right. \\ &\quad + \left(\frac{\partial_F^2 \sigma(F_0) \sigma(F_0)}{6} (2y\phi(y) + 3\bar{\Phi}(y)) \right. \\ &\quad + \left. \frac{(\partial_F \sigma(F_0))^2}{3} y\phi(y) \right) T \\ &\quad + \frac{(\partial_F \sigma(F_0))^2}{4} (y^3 \phi(y) + y\phi(y) + 2\bar{\Phi}(y))T \\ &\quad + \left(\frac{\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) (2y^4 + 3y^2 + 9)}{12} \right. \\ &\quad + \left. \frac{2(\partial_F \sigma(F_0))^3 (y^4 + 3)}{12} \right) T \\ &\quad + \frac{(\partial_F \sigma(F_0))^3}{24} (y^2 - 1)^3 + \frac{\partial_F^3 \sigma(F_0) \sigma^2(F_0)}{12} (y^2 + 3) \\ &\quad + \left. \frac{3\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) \sigma(F_0)}{12} (1 + y^2) \right) \end{aligned}$$

$$\begin{aligned} &+ \frac{\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) \sigma(F_0)}{72} (1 + y^2) \\ &+ \left. \frac{(\partial_F \sigma(F_0))^3}{72} (y^2 - 1) \right) \phi(y) T^{\frac{3}{2}} \Bigg) + \sigma^2(F_0) TE_{QC}(T) \end{aligned} \quad (27)$$

where $G_q^c(y) = (1 + y^2) \bar{\Phi}(y) - y\phi(y)$. The $T^{\frac{3}{2}}$ -order approximation for the price of a quadratic put option is given by:

$$\begin{aligned} \frac{V^{QP}(T, F, K)}{N_T} &= \sigma^2(F_0)T \left(G_q^c(y) - \partial_F \sigma(F_0) \phi(y) \sqrt{T} \right. \\ &\quad - \left(\frac{\partial_F^2 \sigma(F_0) \sigma(F_0)}{6} (2y\phi(y) - 3\Phi(y)) \right. \\ &\quad + \left. \frac{(\partial_F \sigma(F_0))^2}{3} y\phi(y) \right) T \\ &\quad - \frac{(\partial_F \sigma(F_0))^2}{4} (y^3 \phi(y) + y\phi(y) - 2\Phi(y))T \\ &\quad - \left(\frac{\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) (2y^4 + 3y^2 + 9)}{12} \right. \\ &\quad + \left. \frac{2(\partial_F \sigma(F_0))^3 (y^4 + 3)}{12} \right) T \\ &\quad + \frac{(\partial_F \sigma(F_0))^3}{24} (y^2 - 1)^3 \\ &\quad + \frac{\partial_F^3 \sigma(F_0) \sigma^2(F_0)}{12} (y^2 + 3) \\ &\quad + \frac{3\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) \sigma(F_0)}{12} (1 + y^2) \\ &\quad + \left. \frac{\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) \sigma(F_0)}{72} (1 + y^2) \right. \\ &\quad + \left. \frac{(\partial_F \sigma(F_0))^3}{72} (y^2 - 1) \right) \phi(y) T^{\frac{3}{2}} \Bigg) \\ &\quad + \sigma^2(F_0) TE_{QP}(T) \end{aligned} \quad (28)$$

where $G_q^p(y) = (1 + y^2) \Phi(y) + y\phi(y)$.

Finally, the $T^{\frac{3}{2}}$ -order approximation for the price of a quadratic swap price is

$$\begin{aligned} \frac{V^{QS}(T, F, K)}{N_T} &= \sigma^2(F_0)T \\ &\quad \times \left(1 + y^2 + \left(\frac{\partial_F^2 \sigma(F_0) \sigma(F_0)}{2} + \frac{(\partial_F \sigma(F_0))^2}{2} \right) T \right) \end{aligned} \quad (29)$$

Proof With the aim of calculating the integral terms in 5.1, we will first compute the conditional expectation of each term using Appendix 2. Next, with the help of Appendix

4 we will compute $\mathbb{E}^N(\cdot)$ for each of the terms. Finally, if we do $\epsilon \rightarrow 1$, we arrive at the desired price approximation. In addition, the error terms are bounded as $|E_i(T)| \leq CT^2$ through the Malliavin's integration by parts, where $i = QC, QP$ and QS . ■

REMARK 4 The final expressions exhibit the parity condition for quadratic options, specifically

$$V^{QS}(T, F, K) = V^{QP}(T, F, K) + V^{QC}(T, F, K).$$

EXAMPLE 5.4 To verify the accuracy of (27), we apply it to the normal SABR model, i.e. $\beta = 0$. In this case, the equivalent local volatility has a closed-form solution, as presented in Balland and Tran (2013). The expression for the equivalent local volatility is given by:

$$\sigma(K) = \alpha \sqrt{1 + 2\rho v \left(\frac{K - F_0}{\alpha} \right) + v^2 \left(\frac{K - F_0}{\alpha} \right)^2}. \quad (30)$$

We have that

$$\begin{aligned} \partial_F \sigma(F_0) &= \rho v, \\ \partial_F^2 \sigma(F_0) &= \frac{v^2}{\alpha} (1 - \rho^2) = \frac{v^2 \bar{\rho}^2}{\alpha} \\ \partial_F^3 \sigma(F_0) &= 3 \frac{v^3 \rho \bar{\rho}^2}{\alpha^2}. \end{aligned}$$

Substituting the above equalities into the call option expansion (27), we derive the following approximation for $V^{QC}(K)$.

$$\begin{aligned} & \frac{V^{QC}(F, K)}{N_T} \\ & \cong \alpha^2 T \left(G_q(y) + \rho v \phi(y) \sqrt{T} + \right. \\ & \quad + \left(\frac{v^2 \rho^2 (y^3 \phi(y) + y \phi(y) + 2 \bar{\Phi}(y))}{4} \right. \\ & \quad + \left. \frac{2v^2 \rho^2 y \phi(y) + v^2 \bar{\rho}^2 (2y \phi(y) + 3 \bar{\Phi}(y))}{6} \right) T \Big) \\ & \quad + \left(\frac{v^3 \bar{\rho}^2 \rho (2y^4 + 3y^2 + 9) \phi(y) + 2v^3 \rho^3 (y^4 + 3) \phi(y)}{12} \right. \\ & \quad + \left. \frac{v^3 \rho^3 (y^2 - 1)^3}{24} \phi(y) \right) T^{\frac{3}{2}} \\ & \quad + \left(\frac{-3v^3 \bar{\rho}^2 \rho (y^2 + 3) \phi(y)}{24} \right. \\ & \quad + \frac{3v^2 \bar{\rho}^2 (1 + y^2) \phi(y)}{24} + \frac{v^3 \bar{\rho}^2 \rho (1 + y^2) \phi(y)}{72} \\ & \quad + \left. \frac{v^3 \rho^3 (1 - y^2) \phi(y)}{72} \right) T^{\frac{3}{2}} \Big). \quad (31) \end{aligned}$$

We evaluate the accuracy of (27) by comparing it with the Monte Carlo method and the approximation proposed in Hagan *et al.* (2019) across a range of strikes and maturities. Table 2 shows the SABR parameters used in the simulation,

Table 2. Normal SABR parameters considering an underlying swap tenor of 5 years.

SABR Parameters	2Y	5Y	10Y
α	0.007610	0.007274	0.006867
v	0.3687	0.2944	0.2007
ρ	0.2549	0.2843	0.3104

calibrated to the swaption market for a 5-year underlying swap.

In this analysis, we evaluate the numerical performance of the Watanabe approximation in comparison to Hagan's approximation and a Monte Carlo simulation. The analysis considers both absolute price differences and relative price errors. The relative error, with Monte Carlo prices as the benchmark, measures the normalized deviation of the approximated prices from the Monte Carlo results. In other words, the relative error for Hagan's approximation is defined as $\left| \frac{\text{Price Hagan}}{\text{Price MC}} - 1 \right|$. The Monte Carlo simulations were conducted with 300,000 paths and 50 time steps per year, employing antithetic sampling and variance matching techniques for improved accuracy. The results of this comparison are illustrate in figures 4–6.

In figures 4–6, we notice that the price comparisons across the three methodologies are similar. We use the Monte Carlo method as a benchmark, with an estimated simulated error of 5×10^{-6} approximately, based on the standard deviation of the simulations.

As shown in all three figures, Hagan's and Watanabe's approximations display comparable performance, with similar error magnitudes. Furthermore, both approximations perform better on the left side of the tail than on the right.

While both approximations behave similarly overall, Hagan's approximation shows a noticeable improvement in accuracy around the strike values of 0.03 and 0.035 for the 5Y and 10Y cases, respectively.

We conclude that the two approximations are equivalent, with Watanabe's approximation performing as effectively as the industry standard. While it may not seem to surpass industry standards at first glance, its key advantage lies in its flexibility. This allows it to be applied to a wide variety of payoffs and models, all while maintaining the same level of accuracy as the industry standard.

5.2. Quadratic option under stochastic local volatility

SLV-type models are commonly employed in the industry; see, for example, Henry-Labordère (2005) and Bang (2019). These models facilitate the control of the backbone through a layer of local volatility and its evolution over time. In the interest rate market, the SABR dynamic is widely considered the market standard. We will consider a general SABR dynamics defined by:

$$\begin{aligned} dF_t &= C(F_t) \sigma_t dW_t, \\ d\sigma_t &= v \sigma_t dB_t. \end{aligned} \quad (32)$$

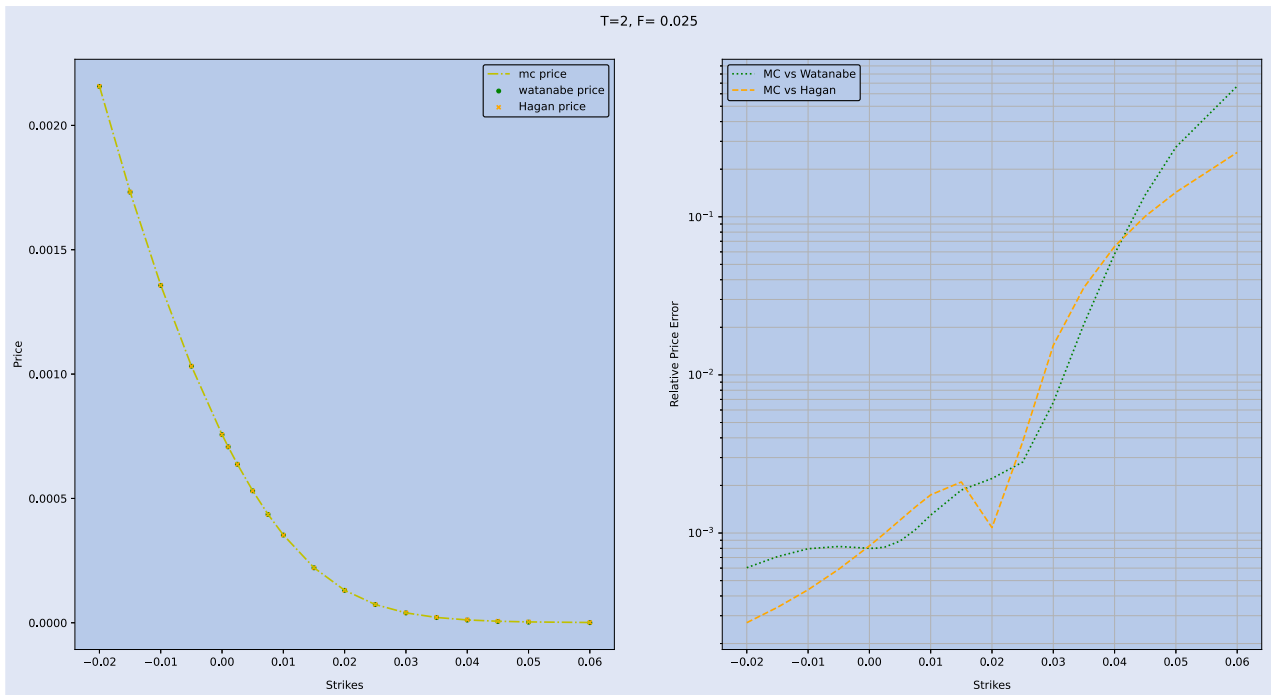


Figure 4. Comparison of call options using Watanabe's expansion against the industry standards. The parameters considered are $\alpha = 0.007610$, $v = 0.3687$, $\rho = 0.2549$, and $T = 2Y$.

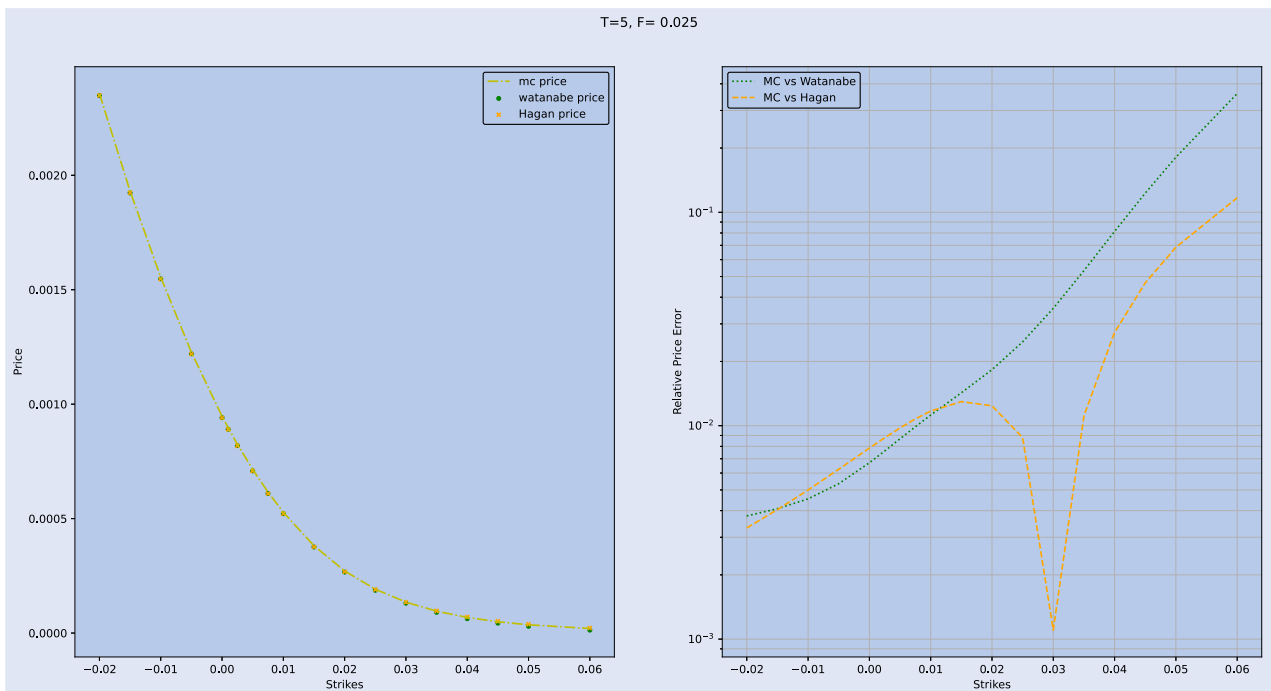


Figure 5. Comparison of call options using Watanabe's expansion against the industry standards. The parameters considered are $\alpha = 0.007274$, $v = 0.2944$, $\rho = 0.2843$, and $T = 5Y$.

Here, $\langle dW_t, dB_t \rangle = \rho dt$ and $C(\cdot)$ is a positive function that belongs to $\mathcal{C}_b^3(\mathbb{R})$. To derive an expansion for F_t analogous to (25), we will consider the following parametrization of the above dynamic

$$\begin{aligned} dF_{\epsilon,t} &= \epsilon C(F_{\epsilon,t}) \sigma_{\epsilon,t} dW_t, \\ d\sigma_{\epsilon,t} &= \epsilon v \sigma_{\epsilon,t} dB_t. \end{aligned} \quad (33)$$

LEMMA 5.5 *Under the dynamics defined by a general SLV model as described in (33), and under the assumption of regularity of $C(\cdot)$, the forward price dynamics can be expressed as follows:*

$$\begin{aligned} F_{\epsilon,t} &= F_0 + \epsilon \sigma(F_0) I_{(1)}(T) \\ &\quad + \epsilon^2 \left(\partial_F C(F_0) \alpha I_{(1,1)}(T) + v \int_0^T \int_0^{u_1} dB_{u_2} dW_{u_1} \right) \end{aligned}$$

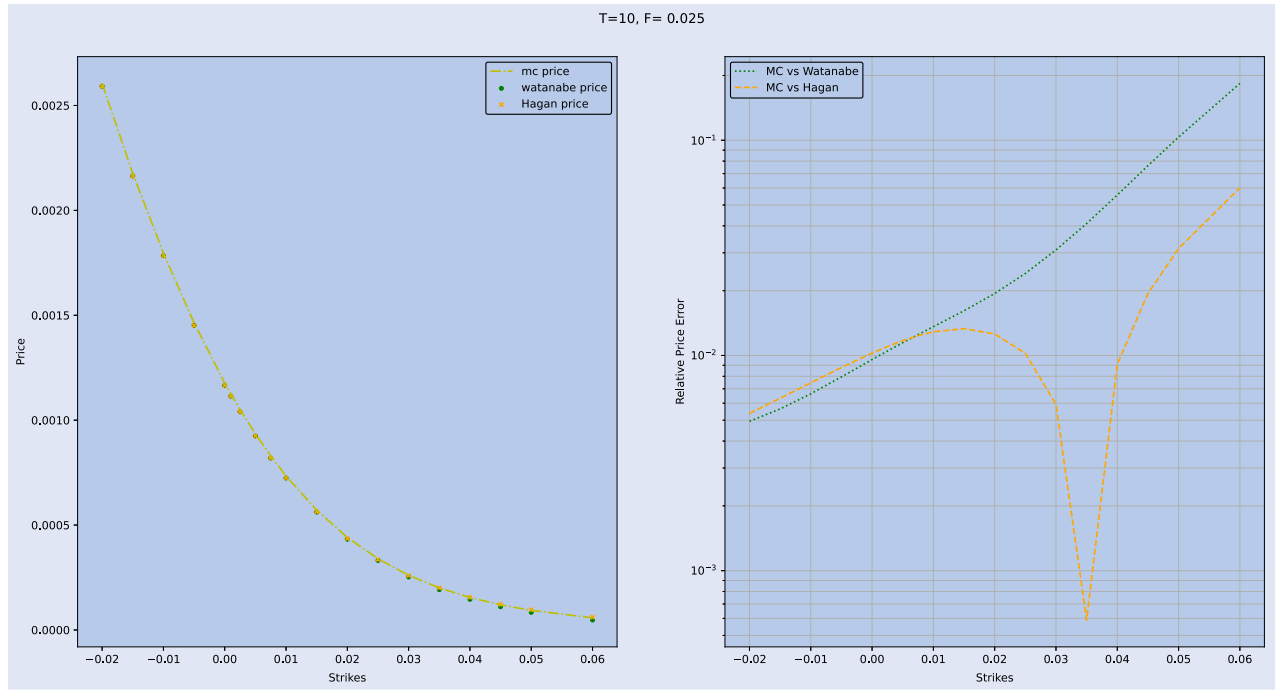


Figure 6. Comparison of call options using Watanabe's expansion against the industry standards. The parameters considered are $\alpha = 0.006897$, $v = 0.2007$, $\rho = 0.3104$, and $T = 10Y$.

$$\begin{aligned}
& + \epsilon^3 \left(\frac{\partial_F^2 C(F_0) C^2(F_0) \alpha^3}{2} \int_0^T W_u^2 dW_u \right. \\
& + 3 \partial_F C(F_0) \alpha \int_0^T F_{2,u} dW_u \\
& + \partial_F C(F_0) C(F_0) v \alpha^2 \int_0^T B_u W_u dW_u \\
& + \frac{v^2 \alpha C(F_0)}{2} \int_0^T \int_0^u \int_0^{u_1} dB_{u_2} dB_{u_1} dW_u \Big) \\
& + \epsilon^4 \left(\alpha \partial_\epsilon^3 C(F_{u,\epsilon})|_{\epsilon=0} + C(F_0) \sigma_{3,u} \right. \\
& + 3 \sigma_{2,u} \partial_\epsilon C(F_{u,\epsilon})|_{\epsilon=0} + 3 \sigma_{1,u} \partial_\epsilon^2 C(F_{u,\epsilon})|_{\epsilon=0} \Big) \\
& + \mathcal{O}(\epsilon^4)
\end{aligned} \tag{34}$$

Proof The result is straightforward by iteratively applying the Itô-Taylor expansion. ■

Therefore, as previously established, we can deduce the terms $\hat{g}_i$.

$$\begin{aligned}
\hat{g}_1(T) &= \frac{W_T}{\sqrt{T}} \\
\hat{g}_2(T) &= \partial_F C(F_0) \frac{\alpha}{\sqrt{T}} I_{(1,1)}(T) + \frac{v}{\sqrt{T}} \int_0^T \int_0^u dB_{u_2} dW_{u_1} \\
\hat{g}_3(T) &= \frac{\partial_F^2 C(F_0) C(F_0) \alpha^2}{2\sqrt{T}} \int_0^T W_u^2 dW_u \\
&+ \partial_F C(F_0) \alpha \int_0^T \hat{g}_2(u) dW_u \\
&+ \frac{\partial_F C(F_0) v \alpha}{\sqrt{T}} \int_0^T B_u W_u dW_u
\end{aligned}$$

$$\begin{aligned}
& + \frac{v^2}{2\sqrt{T}} \int_0^T \int_0^u \int_0^{u_1} dB_{u_2} dB_{u_1} dW_u, \\
\hat{g}_4(T) &= \frac{1}{6\alpha C(F_0) \sqrt{T}} \int_0^T \left(\sigma_0 \partial_\epsilon^3 C(F_{u,\epsilon})|_{\epsilon=0} + C(F_0) \sigma_{3,u} \right. \\
&+ 3 \sigma_{2,u} \partial_\epsilon C(F_{u,\epsilon})|_{\epsilon=0} \\
&+ 3 \sigma_{1,u} \partial_\epsilon^2 C(F_{u,\epsilon})|_{\epsilon=0} \Big) dW_u.
\end{aligned} \tag{35}$$

THEOREM 5.6 (Stochastic local volatility expansion for quadratic payoffs) *Under the model specified by (33), let y be defined as in (11), where $\phi(\cdot)$ is standard normal pdf, $\Phi(\cdot)$ the standard normal cdf, and $\bar{\Phi}(\cdot) = 1 - \Phi(\cdot)$.*

Then, the T_2^3 -order quadratic call option approximation price is

$$\begin{aligned}
& \frac{V^{QC}(T, F, K)}{N_T} \\
&= \alpha^2 C^2(F_0) T \left(G_q(T) + (v\rho + \alpha \partial_F C(F_0)) \phi(y) \sqrt{T} \right. \\
&+ \left(\frac{(\partial_F C(F_0))^2 + 2 \partial_F C(F_0) \rho v + v^2 \rho^2}{4} \right. \\
&\times (y^3 \phi(y) + y \phi(y) + 2 \bar{\Phi}(y)) \\
&+ \left(\frac{C(F_0) \alpha^2 \partial_F^2 C(F_0) + \partial_F C(F_0) v \alpha \rho + v^2}{6} \right. \\
&\times (2y \phi(y) + 3 \bar{\Phi}(y)) \\
&+ \left. \frac{(\partial_F C(F_0))^2 \alpha + v \rho \alpha \partial_F C(F_0)}{3} y \phi(y) - \frac{\rho^2 v^2}{2} \bar{\Phi}(y) \right) T
\end{aligned}$$

$$\begin{aligned}
& + \left(\partial_F C(F_0) \alpha \left(\left(\frac{C(F_0) \alpha^2 \partial_F^2 C(F_0)}{12} + \partial_F C(F_0) v \alpha \rho + \rho^2 v^2 \right) \right. \right. \\
& \times (2y^2 + 3y^2 + 9) \\
& + \left(\frac{(\partial_F C(F_0))^2 \alpha + v \rho \alpha \partial_F C(F_0)}{6} (y^4 + 3) \right. \\
& \left. \left. - \frac{\rho^2 v^2}{4} (y^2 + 1) \right) \right) \\
& + \frac{(\partial_F C(F_0))^3 \alpha^3 + 3(\partial_F C(F_0))^2 \alpha^2 v \rho + 6 \partial_F C(F_0) \alpha v^2 \rho^2 + v^3 \rho^3}{24} (y^2 - 1)^3 \\
& + \frac{\partial_F C(F_0) \alpha v^2 \rho^2 + 2v^3 \rho \bar{\rho}^2 v^2 \bar{\rho}^2 \rho}{6} (y^2 - 1)(2y^2 + 1) \\
& + \frac{2 \partial_F C(F_0) v^2 \bar{\rho}^2 \alpha + \partial_C(F_0) \alpha v \bar{\rho}^2}{6} \\
& + 3 \frac{\partial_F C(F_0) \rho^2 \alpha v + \alpha^2 (\partial_F C(F_0))^2}{24} (y^2 + 1) \\
& + \frac{\partial_F^3 \sigma(F_0) \sigma^2(F_0)}{12} (y^2 + 3) \\
& + \frac{3 \partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) \sigma(F_0)}{12} (1 + y^2) \\
& + \frac{\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) \sigma(F_0)}{72} (1 + y^2) \\
& + \frac{(\partial_F \sigma(F_0))^3}{72} (y^2 - 1) + \frac{v^3 \rho^3}{12} \Big) \phi(y) T^{\frac{3}{2}} \Big) \\
& + \alpha^2 C^2(F_0) TE_{QC}(T)
\end{aligned} \tag{36}$$

where $G_q^c(y) = (1 + y^2) \bar{\Phi}(y) - y \phi(y)$.

The $T^{\frac{3}{2}}$ -order quadratic put option approximation price is given by

$$\begin{aligned}
& \frac{V^{QP}(T, F, K)}{N_T} \\
& = \alpha^2 C^2(F_0) T \left(G_q(T) - (v \rho + \alpha \partial_F C(F_0)) \phi(y) \sqrt{T} \right. \\
& \quad - \left(\frac{(\partial_F C(F_0))^2 + 2 \partial_F C(F_0) \rho v + v^2 \rho^2}{4} \right. \\
& \quad \times (y^3 \phi(y) + y \phi(y) - 2 \Phi(y)) \\
& \quad - \left(\frac{C(F_0) \alpha^2 \partial_F^2 C(F_0) + \partial_F C(F_0) v \alpha \rho + v^2}{6} \right. \\
& \quad \times (2y \phi(y) - 3 \Phi(y)) \\
& \quad + \frac{(\partial_F C(F_0))^2 \alpha + v \rho \alpha \partial_F C(F_0)}{3} y \phi(y) - \frac{\rho^2 v^2}{2} \Phi(y) \Big) T \\
& \quad - \left(\partial_F C(F_0) \alpha \left(\left(\frac{C(F_0) \alpha^2 \partial_F^2 C(F_0)}{12} + \partial_F C(F_0) v \alpha \rho + \rho^2 v^2 \right) \right. \right. \\
& \quad \times (2y^2 + 3y^2 + 9)
\end{aligned}$$

$$\begin{aligned}
& + \left(\frac{(\partial_F C(F_0))^2 \alpha + v \rho \alpha \partial_F C(F_0)}{6} (y^4 + 3) \right. \\
& \left. - \frac{\rho^2 v^2}{4} (y^2 + 1) \right) \\
& + \frac{(\partial_F C(F_0))^3 \alpha^3 + 3(\partial_F C(F_0))^2 \alpha^2 v \rho + 6 \partial_F C(F_0) \alpha v^2 \rho^2 + v^3 \rho^3}{24} (y^2 - 1)^3 \\
& + \frac{\partial_F C(F_0) \alpha v^2 \rho^2 + 2v^3 \rho \bar{\rho}^2 v^2 \bar{\rho}^2 \rho}{6} (y^2 - 1)(2y^2 + 1) \\
& + \frac{2 \partial_F C(F_0) v^2 \bar{\rho}^2 \alpha + \partial_C(F_0) \alpha v \bar{\rho}^2}{6} \\
& + 3 \frac{\partial_F C(F_0) \rho^2 \alpha v + \alpha^2 (\partial_F C(F_0))^2}{24} (y^2 + 1) \\
& + \frac{\partial_F^3 \sigma(F_0) \sigma^2(F_0)}{12} (y^2 + 3) \\
& + \frac{3 \partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) \sigma(F_0)}{12} (1 + y^2) \\
& + \frac{\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) \sigma(F_0)}{72} (1 + y^2) \\
& + \frac{(\partial_F \sigma(F_0))^3}{72} (y^2 - 1) + \frac{v^3 \rho^3}{12} \Big) \phi(y) T^{\frac{3}{2}} \Big) \\
& + \alpha^2 C^2(F_0) TE_{QP}(T)
\end{aligned} \tag{37}$$

where $G_q^p(y) = (1 + y^2) \Phi(y) + y \phi(y)$.

Finally, the $T^{\frac{3}{2}}$ -order quadratic swap price is given by

$$\begin{aligned}
& \frac{V^{QS}(T, F, K)}{N_T} \\
& = \alpha^2 C^2(F_0) T \left((1 + y^2) \right. \\
& \quad + \left(\frac{(\partial_F C(F_0))^2 + 2 \partial_F C(F_0) \rho v + v^2 \rho^2}{2} - \frac{\rho^2 v^2}{2} \right. \\
& \quad + \frac{C(F_0) \alpha^2 \partial_F^2 C(F_0) + \partial_F C(F_0) v \alpha \rho + v^2}{2} \Big) T \Big) \\
& \quad + \alpha^2 C^2(F_0) TE_{QS}(T).
\end{aligned} \tag{38}$$

Proof We consider the quadratic price expansion presented in Theorem 5.1. Using (A.13), (A.14), and (A.15), we compute $\mathbb{E}^N(|\hat{g}_1^i(T)|)$ for each term. To approximate the price, we must evaluate $\mathbb{E}^N(\cdot)$ for each component. For this purpose, we apply Appendix 3 to the results obtained in the previous step. Following the same approach as before, and under the regularity assumption for the process (33), the error terms are bounded as $|E_i(T)| \leq CT^2$ through Malliavin's integration by part, where $i = QC, QP, QS$. ■

REMARK 5 It is important to note that when $v = 0, \rho = 0$ and $\alpha = 1$, the model (32) simplifies to a pure local volatility model. Substituting these values into (36) exactly recovers (27) up to order T .

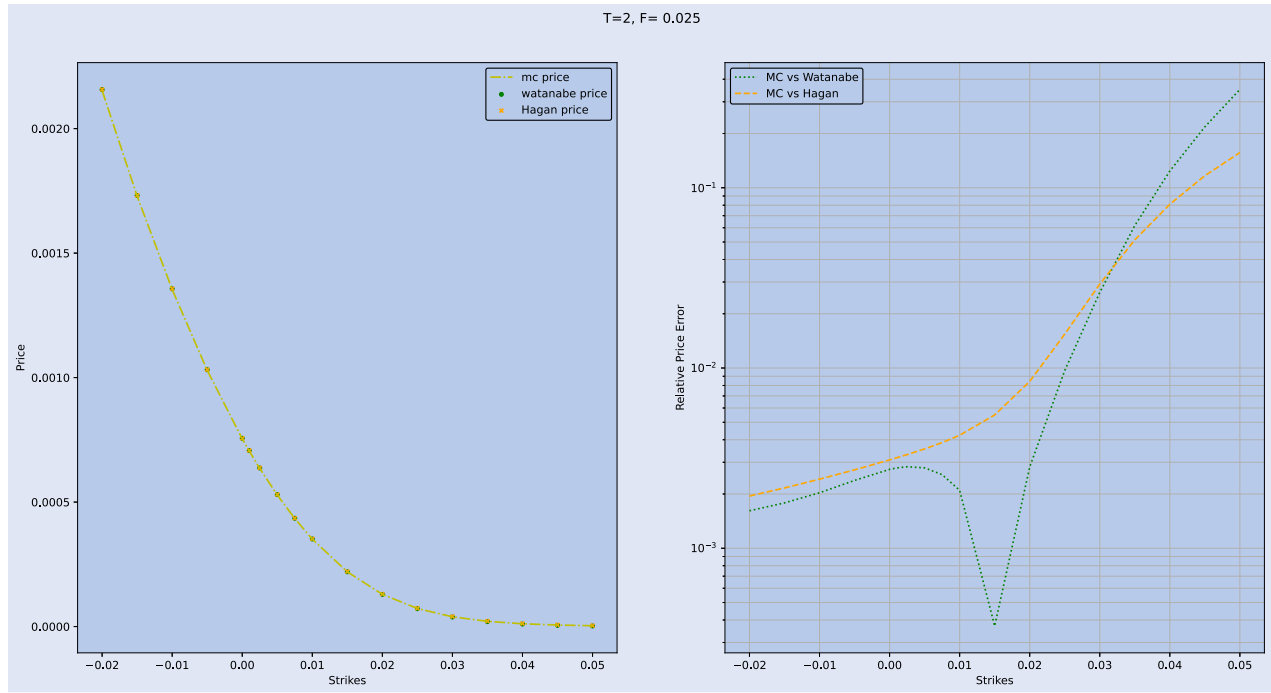


Figure 7. Comparison of call options using Watanabe's expansion against the industry standards. The parameters considered are $\alpha = 0.007610$, $\nu = 0.3687$, $\rho = 0.2549$, and $T = 2Y$.

EXAMPLE 5.7 To verify the accuracy of (36), we assign $C(F) = 1$, reducing the dynamic equation (32) to the standard SABR model. We use the parameter set described in (5.4). In this case, the approximation (36) is reduced to:

$$\begin{aligned} & \frac{V^{QC}(T, F, K)}{N_T} \\ & \cong \alpha^2 t \left(G_q(T) + \rho v \phi(y) \sqrt{T} + \left(\frac{v^2 \rho^2 y \phi(y)}{3} \right. \right. \\ & \quad + \frac{v^2 \rho^2}{4} (y^3 \phi(y) + y \phi(y) + 2 \bar{\Phi}(y)) \\ & \quad + \left. \frac{v^2 \rho^2}{6} (2y \phi(y) + 3 \bar{\Phi}(y)) \right) T \\ & \quad + \left(\frac{m_c(y) \phi(y)}{3} + \frac{v^3 \rho^3}{12} y \phi(y) \right. \\ & \quad + \left. \frac{v^3 \rho^3}{6} (y^4 - y^2 + 1) \phi(y) \right) T^{\frac{3}{2}}, \end{aligned}$$

where

$$m_c(y) = (y^2 - 1) \left(\frac{v^3 \rho^3 (y^2 - 1)^2}{8} + \frac{v^3 \rho^3 (2y^2 + 1)}{4} \right).$$

The analysis considers both absolute price differences and relative price errors. The relative error, with Monte Carlo prices as the benchmark, measures the normalized deviation of the approximated prices from the Monte Carlo results. In other words, the relative error for Hagan's approximation is defined as $\left| \frac{\text{Price Hagan}}{\text{Price MC}} - 1 \right|$. The Monte Carlo simulations were conducted with 300,000 paths and 50 time steps per year, employs a moment-matching-to-maturity technique to reduce computational effort when sampling F_T . The comparison of results is shown in the following figures.

Figures 7–9, show that the price comparisons across the three methodologies are closely aligned. We consider the Monte Carlo method as the benchmark, with an estimated simulated error of approximately 5×10^{-6} , based on the standard deviation of the simulations.

As shown in all three figures, both Hagan's and Watanabe's approximations show consistent performance, with the same error magnitude. Furthermore, both methods produce better results on the left side of the tail compared to the right.

While both approximations perform similarly overall, Watanabe's approximation shows greater accuracy at the strike values of 0.015, -0.05 , and -0.015 for the 2Y, 5Y, and 10Y cases, respectively.

In conclusion, both approximations offer comparable accuracy, with Watanabe's approximation performing at the same level as industry standards. Although it may not initially appear superior, this flexibility allows it to adapt to a wide range of payoffs and models, consistently maintaining the same level of accuracy as industry benchmarks.

6. Future works

In a recent paper Skoufis (2024), the author proposes a closed-form solution for the convexity adjustment in the context of average RFR indexes when the underlying follows a time-dependent SABR model. This issue was previously addressed in Garcia-Lorite and Merino (2023) for the Cheyette model with local volatility using Malliavin calculus. To ensure that the paper is self-contained, let us introduce the basic concepts. Let us define

$$I(T_a, T_b) = \mathbb{E}^{\mathbb{Q}} \left[\frac{e^{\int_{T_a}^{T_b} r_u du} - 1}{T_b - T_a} \right],$$

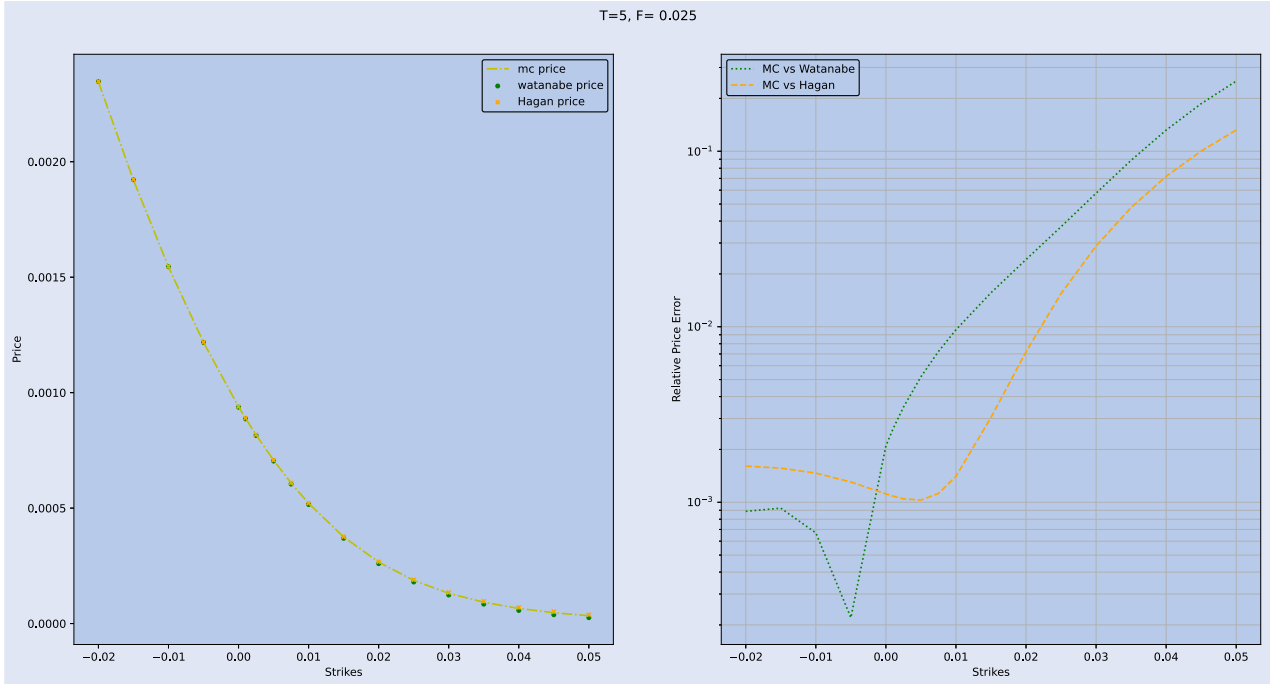


Figure 8. Comparison of call options using Watanabe's expansion against the industry standards. The parameters considered are $\alpha = 0.007274$, $\nu = 0.2944$, $\rho = 0.2843$, and $T = 5Y$.

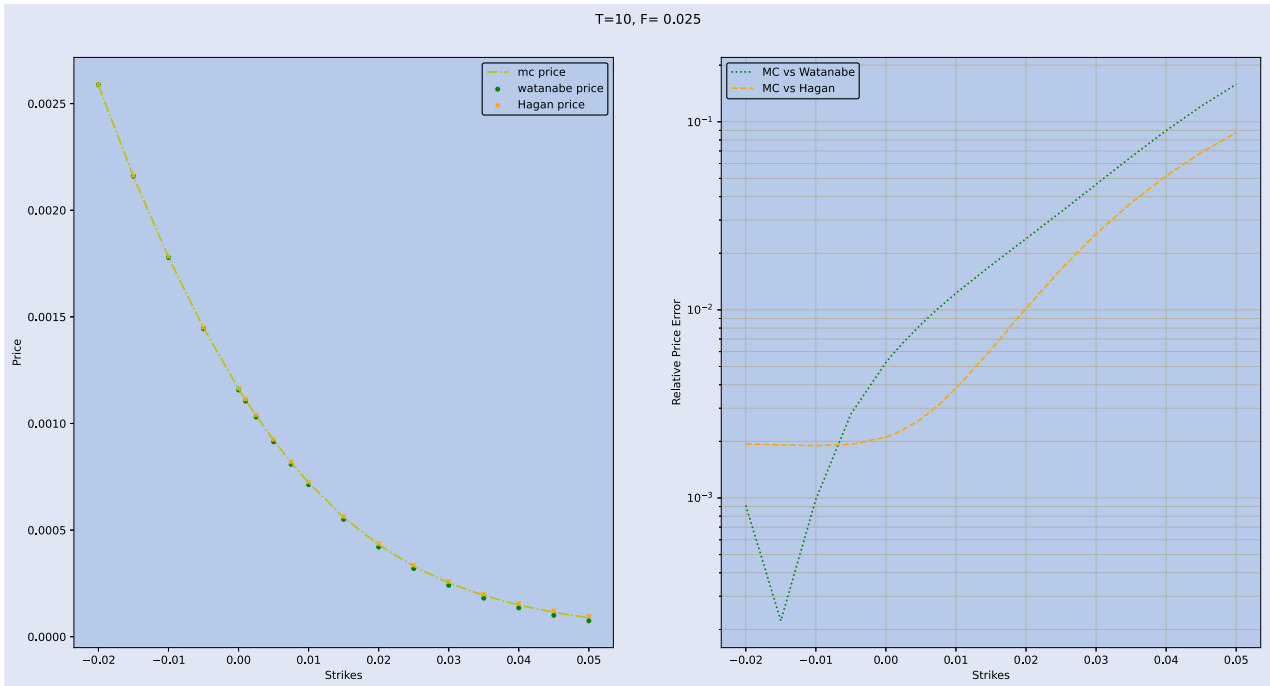


Figure 9. Comparison of call options using Watanabe's expansion against the industry standards. The parameters considered are $\alpha = 0.006897$, $\nu = 0.2007$, $\rho = 0.3104$, and $T = 10Y$.

$$I_{avg}(T_a, T_b) = \frac{\int_{T_a}^{T_b} r_u du}{T_b - T_a},$$

and $R(t) = \mathbb{E}^Q[I(T_a, T_b)]$. In addition, we will suppose that

$$dR_t(T_a, T_b) = h_t C(R_t(T_a, T_b)) \sigma_t dW_t,$$

$$d\sigma_t = \nu \sigma_t dB_t$$

$$\langle dW_t, dB_t \rangle = \rho dt$$

where $h_t = \min \left\{ 1, \left(\frac{T_b - t}{T_b - T_a} \right)^q \right\}$ with $0 < q < 1$. From the definition of $I_{avg}(T_a, T_b)$, we have that

$$\begin{aligned} I_{avg}(T_a, T_a, T_b) &= \frac{\ln(1 + (T_b - T_a)I(T_a, T_b))}{T_b - T_a} \\ &= \frac{\ln(1 + (T_b - T_a)R(T_a, T_a, T_b))}{T_b - T_a}. \end{aligned}$$

Then, the issue is to compute $\mathbb{E}[I_{avg}(T_a, T_b)]$ when R_{t,T_a,T_b} follows the dynamic (6). Using Itô formula, we obtain

(39)

that

$$\begin{aligned} & \mathbb{E}[I_{\text{avg}}(T_a, T_b)] \\ &= I(0, T_a, T_b) \\ &+ \frac{1}{2} \mathbb{E}^{\mathbb{Q}} \left[\int_0^{T_b} G''(R(u, T_a, T_b)) h_u^2 \sigma_u^2 C^2(R(u, T_a, T_b)) ds \right] \\ &\approx I(0, T_a, T_b) + \frac{1}{2} \mathbb{E}^{\mathbb{Q}} G''(R(0, T_a, T_b)) \\ &\times \mathbb{E}^{\mathbb{Q}} [(R(T_b, T_a, T_b) - R(0, T_a, T_b))^2] \end{aligned}$$

where

$$G(R) = \frac{\ln(1 + (T_b - T_a)R)}{T_b - T_a}.$$

Assuming we have an expansion analogous to (9), it follows that

$$\begin{aligned} & \mathbb{E}^{\mathbb{Q}} [(R(T_b, T_a, T_b) - R(0, T_a, T_b))^2] \\ &= \alpha^2 C^2(R(0, T_a, T_b)) \mathbb{E}^{\mathbb{Q}} [\hat{g}_1^2(T_b) + \hat{g}_1^2(T_b) \\ &+ 2\hat{g}_1(T_b)\hat{g}_3(T_b)] T + \mathcal{O}(T^{\frac{3}{2}}). \end{aligned}$$

The objective of future research is to obtain the expansion of $F_{\epsilon,t}$ under a general SLV dynamic. This would enable the computation of the convexity adjustment for an average RFR. Moreover, it would facilitate more efficient pricing of generic terminal payoffs within the framework of a general SLV model.

7. Conclusions

In conclusion, we have proposed a novel approach for pricing CMS derivatives that utilizes Malliavin's calculus and Watanabe's expansions. First, we establish a model-independent connection between the price of a CMS derivative and quadratic payoffs through Malliavin calculus. Next, we extend Watanabe's expansion results to the quadratic payoff case under local and stochastic local volatility. The resulting approximations obtained are generic. To assess their accuracy, we perform a numerical comparison within the SABR model, evaluating our approximations against the industry standards: Hagan's approximation, and a Monte Carlo simulation. Our findings suggest that our approximations are as precise as Hagan's. Moreover, they have the potential to cover a wider range of models and can extend to various payoff structures, thereby providing a flexible framework for pricing different financial derivatives.

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Disclosure statement

The authors declare that there are no conflicts of interest. They are solely responsible for the content and writing of this paper.

References

- Ait-Sahalia, Y., Li, C. and Li, C.X., Implied stochastic volatility models. *Rev. Financ. Stud.*, 2021a, **34**(1), 394–450. <https://doi.org/10.1093/rfs/hhaa041>
- Ait-Sahalia, Y., Li, C. and Li, C.X., Closed-form implied volatility surfaces for stochastic volatility models with jumps. *J. Econom.*, 2021b, **222**(1), 364–392. <https://doi.org/10.1016/j.jeconom.2020.07.006>
- Alòs, E., A decomposition formula for option prices in the Heston model and applications to option pricing approximation. *Finance Stoch.*, 2012, **16**(3), 403–422. <https://doi.org/10.1007/s00780-012-0177-0>
- Alòs, E. and García Lorite, D., *Malliavin Calculus in Finance: Theory and Practice*, 2021 (Chapman and Hall/CRC: New York). <https://doi.org/10.1201/9781003018681>
- Alòs, E., Gatheral, J. and Radoičić, R., Exponentiation of conditional expectations under stochastic volatility. *Quant. Finance*, 2020, **20**(1), 12–27. <https://doi.org/10.1080/14697688.2019.1642506>
- Andersen, L. and Piterbarg, V., *Interest Rate Modeling. Volume 3: Term Structure Models*, 2010 (Atlantic Financial Press: New York & London).
- Balland, P. and Tran, Q., SABR goes normal. *Risk*, 2013, **26**, 72–77.
- Bang, D., Local-stochastic volatility for vanilla modeling: A tractable and arbitrage free approach to option pricing. SSRN, 2019. <https://doi.org/10.2139/ssrn.3171877>.
- Fang, F. and Oosterlee, C.W., A novel pricing method for European options based on Fourier-cosine series expansions. *SIAM J. Sci. Comput.*, 2009, **31**(2), 826–848. <https://doi.org/10.1137/080718061>
- García-Lorite, D. and Merino, R., Convexity adjustments à la Malliavin, 2023. arXiv:2304.13402.
- Gatheral, J., Hsu, E., Laurence, P., Ouyang, C. and Wang, T., Asymptotics of implied volatility in local volatility models. *Math. Finance*, 2012, **22**(4), 591–620. <https://doi.org/10.1111/j.1467-9965.2010.00472.x>
- Gulisashvili, A., Lagunas, M., Merino, R. and Vives, J., Higher order approximation of call option prices under stochastic volatility models. *J. Comput. Finance*, 2020, **24**(1), 1–20. <https://doi.org/10.21314/JCF.2020.387>
- Hagan, P., Convexity conundrums: Pricing cms swaps, caps and floors. *Wilmott*, 2003, **2003**(2), 38–45. <https://doi.org/10.1002/wilm.42820030211>
- Hagan, P. and Woodward, D., An end to replication. *Risk*, May 2021. <https://www.risk.net/media/download/1063311/download>
- Hagan, P., Lesniewski, A., Skoufis, G.E. and Woodward, D., Explicit pricing of quadratic derivatives under SABR. *Wilmott*, 2019, **2019**(103), 40–52. <https://doi.org/10.1002/wilm.10792>
- Hagan, P.S. and Woodward, D.E., Equivalent Black volatilities. *Appl. Math. Finance*, 1999, **6**(3), 147–157.
- Hagan, P.S., Kumar, D., Lesniewski, A. and Woodward, D.E., Managing smile risk. *Wilmott Mag.*, 2002, **2002**(1), 84–108.
- Henry-Labordère, P., A general asymptotic implied volatility for stochastic volatility models, 2005. Preprint.
- Henry-Labordère, P., *Analysis, Geometry, and Modeling in Finance: Advanced Methods in Option Pricing*, 2008 (Chapman and Hall/CRC: New York). <https://doi.org/10.1201/9781420087000>
- Jacquier, A. and Lorig, M., From characteristic functions to implied volatility expansions. *Adv. Appl. Probab.*, 2015, **47**(3), 837–857. <https://doi.org/10.1239/aap/1444308884>
- Kahl, C., Modelling and simulation of stochastic volatility in finance. Dissertation.com, 2008. Available online at: <https://api.semanticscholar.org/CorpusID:152401201>.
- Karami, Y. and Shiraya, K., An approximation formula for normal implied volatility under general local stochastic volatility models.

- J. Futures Mark.*, 2018, **38**(9), 1043–1061. <https://doi.org/10.1002/fut.21931>
- Kunitomo, N. and Takahashi, A., On validity of the asymptotic expansion approach in contingent claim analysis. *Ann. Appl. Probab.*, 2003, **13**(3), 914–952. <https://doi.org/10.1214/aoap/1060202831>
- Li, C., Maximum likelihood estimation for diffusion processes via closed form density expansions. *Ann. Stat.*, June 2013, **41**(3), 1350–1380. <https://doi.org/10.1214/13-AOS1118>
- Li, C., Closed-form expansion, conditional expectation, and option valuation. *Math. Oper. Res.*, 2014, **39**(2), 487–516. <https://doi.org/10.1287/moor.2013.0613>
- Li, C., An, Y., Chen, D., Lin, Q. and Si, N., Efficient computation of the likelihood expansions for diffusion models. *IIE Trans.*, 2016, **48**(12), 1156–1171. <https://doi.org/10.1080/0740817X.2016.1200201>
- Muroi, Y., Pricing of credit derivatives with the asymptotic expansion approach. *J. Comput. Finance*, 2021, **15**(3), 135–171. <https://doi.org/10.21314/JCF.2012.251>
- Nagami, K., Expansion method for pricing foreign exchange options under stochastic volatility and interest rates. *J. Comput. Finance*, 2021, **25**(2), 29–50.
- Nualart, D., *The Malliavin Calculus and Related Topics*, 2nd ed., 1995 (Springer-Verlag: Berlin).
- Pagliarani, S. and Pascucci, A., Local stochastic volatility with jumps: analytical approximations. *Int. J. Theor. Appl. Finance*, 2013, **16**(08), 1350050. <https://doi.org/10.1142/S0219024913500507>
- Osajima, Y., The asymptotic expansion formula of implied volatility for dynamic SABR Model and FX Hybrid Model. Graduate School of Mathematical Sciences, the University of Tokyo, 2006, Preprint.
- Skoufis, G., Sabr model convexity adjustment for the valuation of an arithmetic average approach to option pricing. SSRN, 2024. <https://doi.org/10.2139/ssrn.4694635>.
- Takahashi, A., Takehara, K. and Toda, M., Application of a high-order asymptotic expansion scheme to long-term currency options. *Int. J. Bus. Finance Res.*, 2011, **5**(3), 87–99.
- Takahashi, A., Takehara, K. and Toda, M., A general computation scheme for a high-order asymptotic expansion method. *Int. J. Theor. Appl. Finance*, 2012, **15**(06), 1250044. <https://doi.org/10.1142/S0219024912500446>
- Takahashi, A. and Yamada, T., An asymptotic expansion with push-down of malliavin weights. *SIAM J. Financ. Math.*, 2012, **3**(1), 95–136. <https://doi.org/10.1137/100807624>
- Takahashi, A. and Yamada, T., On error estimates for asymptotic expansions with Malliavin weights: Application to stochastic volatility model. *Math. Oper. Res.*, 2015, **40**(3), 513–541. <https://doi.org/10.1287/moor.2014.0683>
- Takahashi, A. and Yamada, T., A weak approximation with asymptotic expansion and multidimensional Malliavin weights. *Ann. Appl. Probab.*, April 2016, **26**(2), 818–856. <https://doi.org/10.1214/15-AAP1105>
- Watanabe, S., Analysis of wiener functionals (Malliavin calculus) and its applications to heat kernels. *Ann. Probab.*, 1987, **15**(1), 1–39. <https://doi.org/10.1214/aop/1176992255>
- Yoshida, N., Asymptotic expansion for statistics related to small diffusions. *J. Jpn. Stat. Soc.*, 1992a, **22**, 139–159.
- Yoshida, N., Asymptotic expansions of maximum likelihood estimators for small diffusions via the theory of Malliavin-Watanabe. *Probab. Theory Relat. Fields*, 1992b, **92**(3), 275–311.

Appendices

We will consider W_t , $W_{\perp,t}$ two independent Brownian motions.

Appendix 1. Normal SABR

A.1. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2(T)|\hat{g}_1(T)]$

We have that

$$\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2(T)|\hat{g}_1(T)] = \mathbb{E}^{\mathbb{Q}^N}\left[\frac{\nu}{\sqrt{T}}\left(\rho\int_0^T W_u dW_u + \bar{\rho}\int_0^T W_{\perp,u} dW_u\right)\middle|\hat{g}_1(T)\right].$$

Therefore,

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2(T)|\hat{g}_1(T)] &= \frac{\rho\nu}{\sqrt{T}}\mathbb{E}^{\mathbb{Q}^N}\left[\int_0^T W_u dW_u\middle|\hat{g}_1(T)\right] \\ &= \frac{1}{2}\rho\nu\sqrt{T}\left(\hat{g}_1^2(T) - 1\right).\end{aligned}\quad (\text{A1})$$

A.2. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_3(T)|\hat{g}_1(T)]$

We will follow the same approach as in the previous section. Therefore, we have

$$\begin{aligned}\hat{g}_3(T) &= \frac{\nu^2}{2\sqrt{T}}\int_0^T B_u^2 - u dW_u \\ &= \frac{\nu^2}{2\sqrt{T}}\int_0^T B_u^2 dW_u - \frac{\nu^2}{2\sqrt{T}}\int_0^T u dW_u.\end{aligned}$$

We have that

$$\begin{aligned}\frac{\rho^2\nu^2}{2\sqrt{T}}\mathbb{E}^{\mathbb{Q}^N}\left[\int_0^T W_u^2 dW_u\middle|W_T = \hat{g}_1(T)\sqrt{T}\right] \\ = \frac{1}{12}\rho^2\nu^2\left(2\hat{g}_1^3(T) - 3\hat{g}_1(T)\right)T.\end{aligned}\quad (\text{A2})$$

$$\frac{\bar{\rho}^2\nu^2}{2\sqrt{T}}\mathbb{E}^{\mathbb{Q}^N}\left[\int_0^T W_{u,\perp}^2 dW_u\middle|W_T = \hat{g}_1(T)\sqrt{T}\right] = \frac{1}{4}\bar{\rho}^2\nu^2\hat{g}_1(T)T.\quad (\text{A3})$$

Therefore, when combine it, we get that

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_3(T)|\hat{g}_1(T)] &= \frac{1}{12}\rho^2\nu^2\left(2\hat{g}_1^3(T) - 3\hat{g}_1(T)\right)T \\ &\quad - \frac{\rho^2\nu^2}{4}\hat{g}_1(T)T.\end{aligned}\quad (\text{A4})$$

A.3. Calculation of $\mathbb{E}^{\mathbb{Q}^N}(\hat{g}_2^2(T)|\hat{g}_1(T))$

From the definition of $\hat{g}_2(T)$, we have that

$$\begin{aligned}\frac{\nu^2}{T}\left(\underbrace{\rho^2\left(\int_0^T W_u dW_u\right)^2}_{A(T)} + \underbrace{\bar{\rho}^2\left(\int_0^T W_{\perp,u} dW_u\right)^2}_{B(T)}\right. \\ \left. + \underbrace{2\rho\bar{\rho}\left(\int_0^T W_{\perp,u} dW_u\right)\left(\int_0^T W_u dW_u\right)}_{C(T)}\right)\end{aligned}$$

Now, we will compute $\mathbb{E}^{\mathbb{Q}^N}[\cdot|\hat{g}_1(T)]$ for each term. The easiest one is $\mathbb{E}^{\mathbb{Q}^N}[C(T)|\hat{g}_1(T)]$, due to the independence of W_T and $W_{\perp,T}$.

We get

$$\mathbb{E}^{\mathbb{Q}^N} [C(T)|\hat{g}_1(T)] = 0.$$

Then, we will compute the rest of the terms

$$\begin{aligned} & \frac{\nu^2 \rho^2}{T} \mathbb{E}^{\mathbb{Q}^N} \left[\left(\int_0^T W_u \, dW_u \right)^2 \middle| W_T = \hat{g}_1(T) \sqrt{T} \right] \\ &= \frac{1}{4} \nu^2 \rho^2 (\hat{g}_1^2(T) - 1)^2 T. \end{aligned} \quad (\text{A5})$$

For the last term, we must remember the next identity for a Brownian bridge X_s with $X_0 = 0$ and $X_T = \hat{g}_1(T) \sqrt{T}$

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}^N} [dX_u \, dX_s] &= \frac{\hat{g}_1^2(T)}{T} + \frac{\min(u, s)}{t(t - \min(u, s))} + \delta(u - s) \\ &\quad - \frac{1}{(t - \min(u, s))} \, du \, ds. \end{aligned} \quad (\text{A6})$$

Therefore, we have that

$$\begin{aligned} & \frac{\nu^2 \bar{\rho}^2}{T} \mathbb{E}^{\mathbb{Q}^N} \left[\left(\int_0^T W_{u,\perp} \, dW_u \right)^2 \middle| W_T = \hat{g}_1(T) \sqrt{T} \right] \\ &= \frac{\nu^2 \bar{\rho}^2}{T} \left(\int_0^T \int_0^T \min(u_1, u_2) \mathbb{E}^{\mathbb{Q}^N} \right. \\ &\quad \times \left. [dW_{u_1} \, dW_{u_2} \mid W_T = \hat{g}_1(T) \sqrt{T}] \right) \\ &= \frac{\nu^2 \bar{\rho}^2}{T} \int_0^T \int_0^T \min(u_1, u_2) \left(\frac{\hat{g}_1^2(T)t}{t^2} + \frac{\min(u_1, u_2)}{t(t - \min(u_1, u_2))} \right. \\ &\quad \left. + \delta(u_1 - u_2) - \frac{1}{t - \min(u_1, u_2)} \right) \, du_1 \, du_2 \\ &= \frac{1}{6} \nu^2 \bar{\rho}^2 (2\hat{g}_1^2(T) + 1) T. \end{aligned} \quad (\text{A7})$$

Now, if we join (A5) and (A7), we obtain that

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2^2(T) \mid W_T = \hat{g}_1(T) \sqrt{T}] &= \frac{\nu^2 \rho^2}{4} (\hat{g}_1^2(T) - 1)^2 T \\ &\quad + \frac{\nu^2 \bar{\rho}^2}{6} (2\hat{g}_1^2(T) + 1) T \end{aligned} \quad (\text{A8})$$

A.4. Calculation of $\mathbb{E}^{\mathbb{Q}^N} (\hat{g}_4(T) | \hat{g}_1(T))$

Of a way similar to the calculation of $\mathbb{E}^{\mathbb{Q}^N} (\hat{g}_3(T) | \hat{g}_1(T))$ in A.2 we have that

$$\begin{aligned} & \mathbb{E}^{\mathbb{Q}^N} \left(\frac{\nu^3}{\sqrt{T}} \int_0^T \int_0^{u_1} \int_0^{u_2} \int_0^{u_3} \, dB_{u_4} \, dB_{u_3} \, dB_{u_2} \, dW_{u_1} \right) \\ &= \frac{\rho^3 \nu^3}{24} (\hat{g}_1^4(T) - 6\hat{g}_1^2(T) + 3). \end{aligned}$$

A.5. Calculation of $\mathbb{E}^{\mathbb{Q}^N} (\hat{g}_2(T) \hat{g}_3(T) | \hat{g}_1(T))$

From the expression for $\hat{g}_2(T)$ and $\hat{g}_3(T)$ we have that

$$\mathbb{E}^{\mathbb{Q}^N} (\hat{g}_2(T) \hat{g}_3(T) | \hat{g}_1(T)) = A(T) + B(T),$$

where

$$\begin{aligned} A(T) &= \frac{\nu^3 \rho^3}{12} (\hat{g}_1^4(T) - \hat{g}_1^2(T) + 1) \\ B(T) &= \nu^3 \bar{\rho}^2 \rho (\hat{g}_1^3(T) - 3\hat{g}_1^2(T) + 2\hat{g}_1(T)). \end{aligned}$$

A.6. Calculation of $\mathbb{E}^{\mathbb{Q}^N} (\hat{g}_2^3(T) | \hat{g}_1(T))$

Of a similar way to (A.3) we have that

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2^3(T) | \hat{g}_1(T)] &= \mathbb{E}^{\mathbb{Q}^N} \left[\left(\frac{\nu \rho T}{2} (\hat{g}_1^2(T) - 1) \right. \right. \\ &\quad \left. \left. + \nu \bar{\rho} \int_0^T W_{u,\perp} \, dW_u \right)^3 \middle| \hat{g}_1(T) \right]. \end{aligned}$$

Now, if we use Newton's Binomial in the above equation, we get that

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2^3(T) | \hat{g}_1(T)] &= \frac{\nu^3 \rho^3}{8} (\hat{g}_1^2(T) - 1)^3 T^{\frac{3}{2}} \\ &\quad + \frac{\nu^3 \rho^2 \bar{\rho}}{4} (\hat{g}_1^2(T) - 1)(2\hat{g}_1^2(T) + 1) T^{\frac{3}{2}}. \end{aligned}$$

Appendix 2. Local volatility

A.7. Calculation of $\mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2(T) | \hat{g}_1(T)]$

From the expression for $\hat{g}_1(T)$ in (26) we have

$$\mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2(T) | \hat{g}_1(T)] = \sqrt{T} \frac{\partial_F \sigma(F_0)}{2} (\hat{g}_1^2(T) - 1).$$

A.8. Calculation of $\mathbb{E}^{\mathbb{Q}^N} [\hat{g}_3(T) | \hat{g}_1(T)]$

We have that

$$\mathbb{E}^{\mathbb{Q}^N} [\hat{g}_3(T) | \hat{g}_1(T)] = A(T) + B(T)$$

where

$$\begin{aligned} A(T) &= \frac{\partial_F^2 \sigma(F_0)}{12} \sigma(F_0) (2\hat{g}_1^3(T) - 3\hat{g}_1(T)) T \\ B(T) &= \frac{(\partial_F \sigma(F_0))^2}{6} H_3(\hat{g}_1(T)) T \end{aligned}$$

where $H_3(x)$ is the Hermite polynomial of order three.

A.9. Calculation of $\mathbb{E}^{\mathbb{Q}^N} [\hat{g}_3(T) \hat{g}_2(T) | \hat{g}_1(T)]$

From the above appendixes, we have that

$$\mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2(T) \hat{g}_3(T) | \hat{g}_1(T)] = \frac{\sqrt{T} (\hat{g}_1^2(T) - 1)}{2} \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_3(T) | \hat{g}_1(T)].$$

Therefore

$$\begin{aligned} & \mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2(T) \hat{g}_3(T) | \hat{g}_1(T)] \\ &= \partial_F \sigma(F_0) \frac{(\hat{g}_1^2(T) - 1)}{24} \left(\partial_F^2 \sigma(F_0) \sigma(F_0) (2\hat{g}_1^3(T) - 3\hat{g}_1(T)) \right. \\ &\quad \left. + 2(\partial_F \sigma(F_0))^2 H_3(\hat{g}_1(T)) \right) T^{\frac{3}{2}} \end{aligned}$$

A.10. Calculation of $\mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2^2(T) | \hat{g}_1(T)]$

From (A.7), we have that

$$\mathbb{E}^{\mathbb{Q}^N} [\hat{g}_2^2(T) | \hat{g}_1(T)] = (\partial_F \sigma(F_0))^2 \frac{(\hat{g}_1^2(T) - 1)^2}{4} T.$$

A.11. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2^3(T)|\hat{g}_1(T)]$

From (A.7), we have that

$$\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2^3(T)|\hat{g}_1(T)] = (\partial_F \sigma(F_0))^3 \frac{(\hat{g}_1^2(T) - 1)^3}{8} T^{\frac{3}{2}}.$$

A.12. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_4(T)|\hat{g}_1(T)]$

From the expression for $\hat{g}_4(T)$ and using the properties of Itô iterated integral, we have that

$$\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_4(T)|\hat{g}_1(T)] = A(T) + B(T) + C(T),$$

where

$$\begin{aligned} A(T) &= \frac{\partial_F^3 \sigma(F_0) \sigma^2(F_0)}{24} (\hat{g}_1^4(T) - 2\hat{g}_1^2(T) - 1), \\ B(T) &= \frac{3\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) \sigma(F_0)}{24} (\hat{g}_1^4(T) - 4\hat{g}_1^2(T) + 1), \\ C(T) &= \frac{\sigma(F_0) \partial_F^2 \sigma(F_0) \partial_F \sigma(F_0)}{144} (3\hat{g}_1^4(T) - 10\hat{g}_1^2(T) + 1) \\ &\quad + \frac{(\partial_F \sigma(F_0))^3}{144} ((\hat{g}_1^4(T) - 6\hat{g}_1^2(T) + 3)). \end{aligned}$$

Appendix 3. Stochastic Local volatility

A.13. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2(T)|\hat{g}_1(T)]$

From (35), we have that

$$\begin{aligned} \mathbb{E}^N[\hat{g}_2(T)|\hat{g}_1(T)] &= \partial_F C(F_0) \mathbb{E}^N\left[\frac{I_{(1,1)}(T)}{\sqrt{T}} | \hat{g}_1(T)\right] \\ &\quad + \nu \rho \mathbb{E}^N\left[\frac{I_{(1,1)}(T)}{\sqrt{T}} | \hat{g}_1(T)\right] \end{aligned}$$

Now, if we use (A6), we have that

$$\mathbb{E}^N[\hat{g}_2(T)|\hat{g}_1(T)] = \frac{\sqrt{T}}{2} (\hat{g}_1^2(T) - 1) (\nu \rho + \alpha \partial_F C(F_0)).$$

A.14. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_3(T)|\hat{g}_1(T)]$

From the expression of $\hat{g}_3(T)$ in (35), we obtain that

$$\mathbb{E}^N[\hat{g}_3(T)|\hat{g}_1(T)] = A(T) + B(T) + C(T) + D(T)$$

where

$$\begin{aligned} A(T) &= C(F_0) \alpha^2 \frac{\partial_F^2 C(F_0)}{2\sqrt{T}} \int_0^T W_u^2 dW_u, \\ B(T) &= \partial_F C(F_0) \alpha \int_0^T \hat{g}_2(u) dW_u, \\ C(T) &= \frac{\partial_F C(F_0) \nu \alpha}{\sqrt{T}} \int_0^T B_u W_u dW_u, \\ D(T) &= \frac{\nu^2}{2\sqrt{T}} \int_0^T \int_0^u \int_0^{u_1} dB_{u_2} dB_{u_1} dW_u. \end{aligned}$$

Through a direct computation and employing the representation of the distinct iterated integrals and the Appendix 1, we get that

$$\mathbb{E}^N[A(T)|\hat{g}_1(T)] = C(F_0) \alpha^2 \frac{\partial_F^2 C(F_0)}{12} (2\hat{g}_1^3(T) - 3\hat{g}_1(T))T,$$

$$\begin{aligned} \mathbb{E}^N[B(T)|\hat{g}_1(T)] &= \frac{(\partial_F C(F_0))^2 \alpha + \nu \rho \alpha \partial_F C(F_0)}{6} \\ &\quad \times (\hat{g}_1^3(T) - 3\hat{g}_1(T))T, \\ \mathbb{E}^N[C(T)|\hat{g}_1(T)] &= \frac{\partial_F C(F_0) \nu \alpha \rho}{12} (2\hat{g}_1^3(T) - 3\hat{g}_1(T))T, \\ \mathbb{E}^N[D(T)|\hat{g}_1(T)] &= \frac{1}{12} \rho^2 \nu^2 (2\hat{g}_1^3(T) - 3\hat{g}_1(T))T \\ &\quad - \frac{\rho^2 \nu^2}{4} \hat{g}_1(T)T. \end{aligned}$$

Rearranging the previous equalities, we obtain that

$$\begin{aligned} \mathbb{E}^N[\hat{g}_3(T)|\hat{g}_1(T)] &= \left(\frac{C(F_0) \alpha^2 \partial_F^2 C(F_0) + \partial_F C(F_0) \nu \alpha \rho + \rho^2 \nu^2}{12} \right) \\ &\quad \times (2\hat{g}_1^3(T) - 3\hat{g}_1(T))T \\ &\quad + \left(\frac{(\partial_F C(F_0))^2 \alpha + \nu \rho \alpha \partial_F C(F_0)}{6} \right) \\ &\quad \times (\hat{g}_1^3(T) - 3\hat{g}_1(T)) - \frac{\rho^2 \nu^2}{4} \hat{g}_1(T) \Big) T \end{aligned}$$

A.15. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2^2(T)|\hat{g}_1(T)]$

From (A.7) and (A8), we have that

$$\hat{g}_2^2(T) = A(T) + B(T) + C(T)$$

where

$$\begin{aligned} A(T) &= \frac{(\partial_F C(F_0))^2}{T} \left(\int_0^T W_u dW_u \right)^2, \\ B(T) &= \frac{2\partial_F C(F_0) \nu \rho}{T} \int_0^T W_u dW_u \int_0^T B_u dW_u, \\ C(T) &= \frac{\nu^2}{T} \left(\int_0^T B_u dW_u \right)^2. \end{aligned}$$

Then, if we take $\mathbb{E}^N(\cdot|\hat{g}_1(T))$, and we use the properties of the iterated integral we get that

$$\begin{aligned} \mathbb{E}^N[\hat{g}_2^2(T)|\hat{g}_1(T)] &= \left(\frac{(\partial_F C(F_0))^2 + 2\partial_F C(F_0) \rho \nu + \nu^2 \rho^2}{4} \right) \\ &\quad \times (\hat{g}_1^2(T) - 1)^2 \\ &\quad + \frac{\nu^2 \rho^2}{6} (2\hat{g}_1^2(T) + 1) \Big) T \end{aligned}$$

A.16. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2^3(T)|\hat{g}_1(T)]$

From (A.7) and (A8), we have that

$$\hat{g}_2^3(T) = A(T) + B(T) + C(T) + D(T)$$

where

$$\begin{aligned} A(T) &= \frac{(\partial_F C(F_0))^3}{8} \alpha^3 (\hat{g}_1^2(T) - 1)^3 T^{\frac{3}{2}} \\ B(T) &= \frac{3(\partial_F C(F_0))^2 \alpha^2}{4} (\hat{g}_1^2(T) - 1)^2 \sqrt{T} \nu \int_0^T \int_0^u dB_{u_1} dW_u \\ C(T) &= \frac{3\partial_F C(F_0)}{\sqrt{T}} (\hat{g}_1^2(T) - 1) \nu^2 \left(\int_0^T \int_0^u dB_{u_1} dW_u \right)^2 \end{aligned}$$

$$D(T) = \frac{\nu^3}{T^{\frac{3}{2}}} \left(\int_0^T \int_0^u dB_{u_1} dW_u \right)^3.$$

Now, if we take $\mathbb{E}^{\mathbb{Q}^N}[\cdot|\hat{g}_1(T)]$ on each one of the above terms, we have that

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}^N}[A(T)|\hat{g}_1(T)] &= \frac{(\partial_F C(F_0))^3}{8} \alpha^3 (\hat{g}_1^2(T) - 1)^3 T^{\frac{3}{2}}, \\ \mathbb{E}^{\mathbb{Q}^N}[B(T)|\hat{g}_1(T)] &= \frac{3(\partial_F C(F_0))^2 \alpha^2}{8} \nu \rho (\hat{g}_1^2(T) - 1)^3 T^{\frac{3}{2}}, \\ \mathbb{E}^{\mathbb{Q}^N}[C(T)|\hat{g}_1(T)] &= \frac{3\partial_F C(F_0) \alpha \nu^2 \rho^2 (\hat{g}_1^2(T) - 1)^3 T^{\frac{3}{2}}}{4} \\ &\quad + \frac{\partial_F C(F_0) \alpha \nu^2 \bar{\rho}^2}{2} (\hat{g}_1^2(T) - 1) \\ &\quad \times (2\hat{g}_1^2(T) + 1) T^{\frac{3}{2}}, \\ \mathbb{E}^{\mathbb{Q}^N}[D(T)|\hat{g}_1(T)] &= \frac{\nu^3 \rho^3}{8} (\hat{g}_1^2(T) - 1)^3 T^{\frac{3}{2}} \\ &\quad + \nu^3 \rho \bar{\rho}^2 (\hat{g}_1^2(T) - 1) (2\hat{g}_1^2(T) + 1) T^{\frac{3}{2}}.\end{aligned}$$

Then, if we join the previous equation, we have that

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2^2(T)|\hat{g}_1(T)] &= \frac{(\partial_F C(F_0))^3 \alpha^3 + 3(\partial_F C(F_0))^2 \alpha^2 \nu \rho}{8} \\ &\quad \times (\hat{g}_1^2(T) - 1)^3 T^{\frac{3}{2}} \\ &\quad + \frac{\partial_F C(F_0) \alpha \nu^2 \bar{\rho} + 2\nu^3 \bar{\rho}^2 \rho}{2} \\ &\quad \times (\hat{g}_1^2(T) - 1) (2\hat{g}_1^2(T) + 1) T^{\frac{3}{2}}.\end{aligned}$$

A.17. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_4(T)|\hat{g}_1(T)]$

From (35) we can decompose $\hat{g}_4(T)$ in four terms, i.e

$$\hat{g}_4(T) = A(T) + B(T) + C(T) + D(T),$$

where

$$\begin{aligned}A(T) &= \frac{\alpha \partial_\epsilon^3 C(F_{u,\epsilon})|_{\epsilon=0}}{6\alpha C(F_0)\sqrt{T}} \\ &= \alpha \left(\frac{\partial_F^3 \sigma(F_0) \sigma^2 F_0}{24\sqrt{T}} \left(\int_0^T \frac{W_u^4}{4} - \frac{3}{2} \int_0^T W_u^2 dW_u \right) \right. \\ &\quad \left. + \frac{\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) C(F_0)}{8\sqrt{T}} \int_0^T (W_u^3 - W_u) dW_u \right), \\ B(T) &= C(F_0) \frac{\sigma_{3,u}}{6\alpha C(F_0)\sqrt{T}} \\ &= \frac{\nu^3}{\sqrt{T}} \int_0^T \int_0^{u_1} \int_0^{u_2} \int_0^{u_3} dB_{u_4} dB_{u_3} dB_{u_2} dW_{u_1}, \\ C(T) &= \frac{\sigma_{2,u} \partial_\epsilon C(F_{u,\epsilon})|_{\epsilon=0}}{2\alpha C(F_0)\sqrt{T}} = \frac{\nu^2 \alpha \partial_F C(F_0)}{2\sqrt{T}} \int_0^T (B_u^2 - u) W_u dW_u, \\ D(T) &= \frac{\sigma_{1,u} \partial_\epsilon^2 C(F_{u,\epsilon})|_{\epsilon=0}}{2\alpha C(F_0)\sqrt{T}} \\ &= \frac{\alpha^2 \partial_F^2 C(F_0) C(F_0)}{2\sqrt{T}} \int_0^T W_u^2 B_u dW_u \\ &\quad + \frac{\alpha \partial_F C(F_0)}{2} \int_0^T \hat{g}_2(u) dW_u.\end{aligned}$$

Now, if we take $\mathbb{E}^{\mathbb{Q}^N}[\cdot|\hat{g}_1(T)]$ on each one of the above terms, we have that

$$\begin{aligned}\mathbb{E}^N[A(T)|\hat{g}_1(T)] &= \frac{\partial_F^3 \sigma(F_0) \sigma^2(F_0)}{24} (\hat{g}_1^4(T) - 2\hat{g}_1^2(T) - 1) \\ &\quad + \frac{3\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0) C(F_0)}{24} \\ &\quad \times (\hat{g}_1^4(T) - 4\hat{g}_1^2(T) + 1) \\ &\quad + \frac{\partial_F^2 \sigma(F_0) \partial_F \sigma(F_0)}{144} (3\hat{g}_1^4(T) - 10\hat{g}_1^2(T) + 1) \\ &\quad + \frac{(\partial_F \sigma(F_0))^3}{144} (\hat{g}_1^4(T) - 6\hat{g}_1^2(T) + 3), \\ \mathbb{E}^N[B(T)|\hat{g}_1(T)] &= \frac{C(F_0) \rho^3 \nu^3}{24} (\hat{g}_1^4(T) - 6\hat{g}_1^2(T) + 3), \\ \mathbb{E}^N[C(T)|\hat{g}_1(T)] &= \frac{\nu^2 \rho^2 \alpha}{8} \partial_F C(F_0) (\hat{g}_1^4(T) - 2\hat{g}_1^2(T) - 1) \\ &\quad + \frac{\nu^2 \bar{\rho}^2 \alpha}{6} \partial_F C(F_0) (\hat{g}_1^2 - 1), \\ \mathbb{E}^N[D(T)|\hat{g}_1(T)] &= \frac{\alpha^2 (\partial_F C(F_0))^2}{48} (3\hat{g}_1^4(T) - 10\hat{g}_1^2(T) + 1) \\ &\quad + \frac{\partial_F C(F_0) \rho^2 \alpha \nu}{48} (3\hat{g}_1^4(T) - 10\hat{g}_1^2(T) + 1) \\ &\quad + \frac{\partial_C(F_0)}{12} \alpha \nu \bar{\rho}^2 (\hat{g}_1^2(T) - 1).\end{aligned}$$

A.18. Calculation of $\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2(T)\hat{g}_3(T)|\hat{g}_1(T)]$

From the expression for $\hat{g}_2(T)$ and $\hat{g}_3(T)$, we have that

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}^N}[\hat{g}_2(T)\hat{g}_3(T)|\hat{g}_1(T)] &= \frac{\partial_F C(F_0) \alpha}{2} (\hat{g}_1^2(T) - 1) \\ &\quad \times \left(\left(\frac{C(F_0) \alpha^2 \partial_F^2 C(F_0) + \partial_F C(F_0) \nu \alpha \rho + \rho^2 \nu^2}{12} \right) \right. \\ &\quad \cdot (2\hat{g}_1^3(T) - 3\hat{g}_1(T)) \\ &\quad \left. + \left(\frac{(\partial_F C(F_0))^2 \alpha + \nu \rho \alpha \partial_F C(F_0)}{6} (\hat{g}_1^3(T) - 3\hat{g}_1(T)) \right. \right. \\ &\quad \left. \left. - \frac{\rho^2 \nu^2}{4} \hat{g}_1(T) \right) \right) T^{\frac{3}{2}}.\end{aligned}$$

Appendix 4. Gaussian integrals

Throughout this paper, the following integrals will appear repeatedly, these are:

$$\int_y^\infty u^n \phi(u) du, \int_{-\infty}^y u^n \phi(u) du, \int_{-\infty}^\infty u^n \phi(u) du.$$

where $\phi(u)$ is the density function of standard normal variable. From the part-integration and the properties of the standard normal variable we have that

$$\begin{aligned}\int_y^\infty u^n \phi(u) du &= y^{n-1} \phi(y) + n \int_y^\infty u^{n-2} \phi(u) du \\ \int_{-\infty}^y u^n \phi(u) du &= -y^{n-1} \phi(y)\end{aligned}$$

$$\begin{aligned}
& + n \int_{-\infty}^y u_{n-2} \phi(u) \int_{\infty}^y u_{n-2} \phi(u), \\
\int_{-\infty}^{\infty} u^n \phi(u) \, du &= \begin{cases} 0, & n \text{ is odd} \\ (n-1)!!, & n \text{ is even} \end{cases}
\end{aligned}
\qquad
\begin{aligned}
\int_{-\infty}^y \phi(u) \, du &= \Phi(y), \\
\int_y^{\infty} \phi(u) \, du &= \bar{\Phi}(y).
\end{aligned}$$

To be able to conclude the above recurrences, we must know the cases $n = 0$ and $n = 1$. Both case are straightforward

$$\begin{aligned}
\int_y^{\infty} u \phi(u) \, du &= \phi(y), \\
\int_{-\infty}^y u \phi(u) \, du &= -\phi(y)
\end{aligned}$$